

Data Preprocessing Techniques

The background features a large, dark gray circle on the left side, which is partially cut off by the right edge of the frame. To the right of the circle, there is a light blue rectangular area in the top right corner. The main area of the slide is a dark gray color.

Roadmap

- Know Your Data
- Statistical Descriptions of Data
- Data Visualization



The background is a dark gray gradient with a variety of white and light green geometric patterns and shapes. These include circles, squares, rectangles, and 3D blocks, some with dotted or striped textures. There are also wavy lines and concentric circles scattered throughout. A solid orange horizontal bar is positioned at the bottom of the image.

Know Your Data

Types of Data Sets: (1) Record Data

- **Relational records**
 - Relational tables, highly structured

- **Data matrix, e.g. numerical matrix, crosstabs**

	China	England	France	Japan	USA	Total
Active Outdoors Crochet Glove		12.00	4.00	1.00	240.00	257.00
Active Outdoors Lycra Glove		10.00	6.00		323.00	339.00
InFlux Crochet Glove	3.00	6.00	8.00		132.00	149.00
InFlux Lycra Glove		2.00			143.00	145.00
Triumph Pro Helmet	3.00	1.00	7.00		333.00	344.00
Triumph Vertigo Helmet		3.00	22.00		474.00	499.00
Xtreme Adult Helmet	8.00	8.00	7.00	2.00	251.00	276.00
Xtreme Youth Helmet		1.00			76.00	77.00
Total	14.00	43.00	54.00	3.00	1,972.00	2,086.00

- **Transaction data**

TID	Items
1	Bread, Coke, Milk
2	Beer, Bread
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Coke, Diaper, Milk

- **Document data: Term-frequency vector (matrix) of text documents**

Person:

Pers_ID	Surname	First_Name	City
0	Miller	Paul	London
1	Ortega	Alvaro	Valencia
2	Huber	Urs	Zurich
3	Blanc	Gaston	Paris
4	Bertolini	Fabrizio	Rom

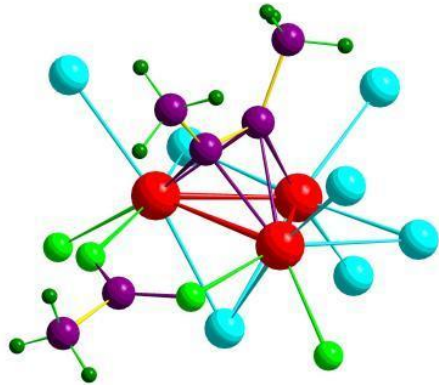
Car_ID	Model	Year	Value	Pers_ID
101	Bentley	1973	100000	0
102	Rolls Royce	1965	330000	0
103	Peugeot	1993	500	3
104	Ferrari	2005	150000	4
105	Renault	1998	2000	3
106	Renault	2001	7000	3
107	Smart	1999	2000	2

no relation

	team	coach	play	ball	score	game	win	lost	timeout	season
Document 1	3	0	5	0	2	6	0	2	0	2
Document 2	0	7	0	2	1	0	0	3	0	0
Document 3	0	1	0	0	1	2	2	0	3	0

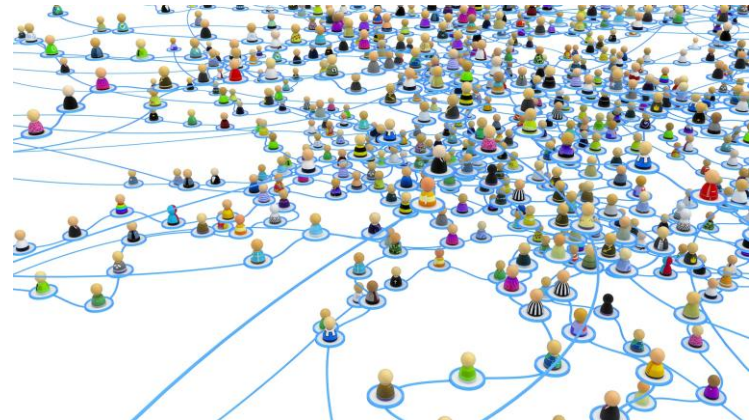
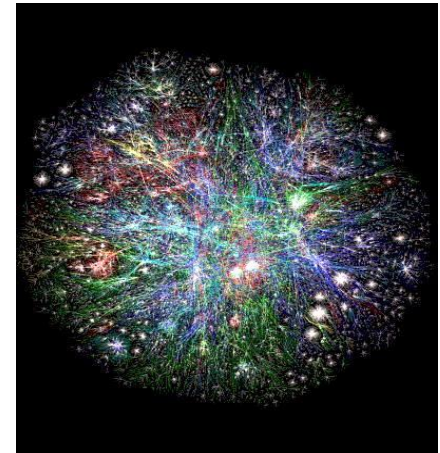
Types of Data Sets: (2) Graphs and Networks

- **Transportation network**
- **World Wide Web**



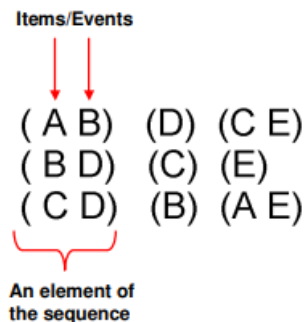
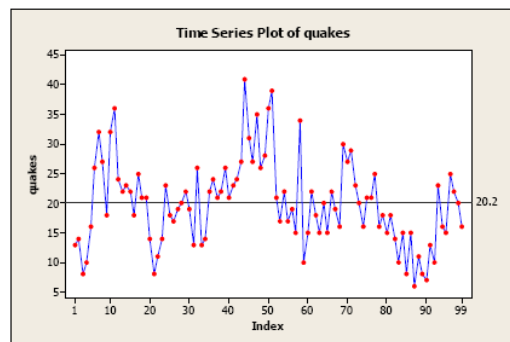
□ **Molecular Structures**

□ **Social or information networks**



Types of Data Sets: (3) Ordered Data

- **Video data:** sequence of images
- **Temporal data:** time-series: For example, a financial data set might contain objects that are time series of the daily prices of various stocks.



- **Sequential Data:** transaction sequences - Time
- **Genetic sequence data (Position) :** the genetic information of plants and animals can be represented in the form of sequences of nucleotides that are known as genes.

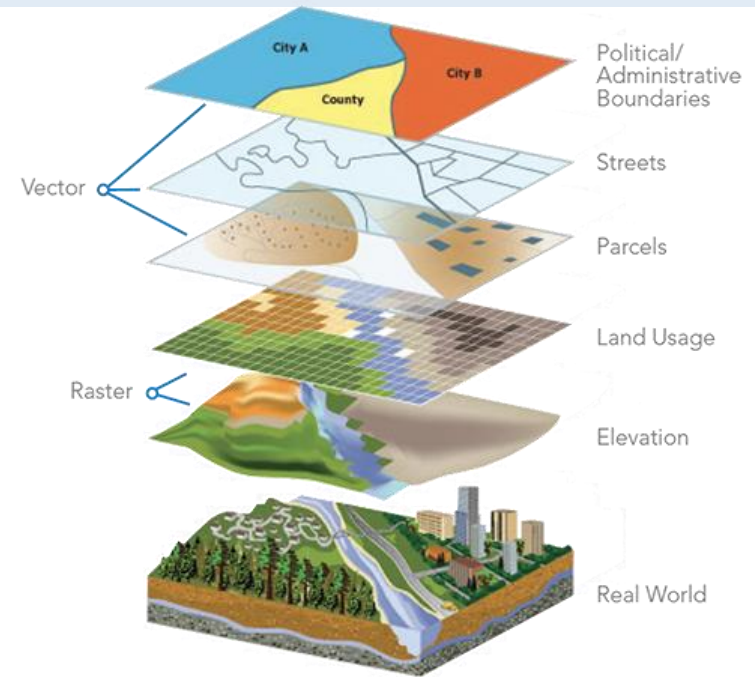


Start

Human	GTTTGGAGG...ATGTTCAACAAATGCTCCTTTTCATTCCTCTATTACAGACCTGCCGCA
Chimpanzee	GTTTGGAGG...ATGTTCAAAATGCTGCTTTCACTCCTCTATTACAGACCTGCCGCA
Macaque	GTTTGGAGG...ATGTTCAAAATGCTCCTTTTCATTCCTCTATTACAGCTGCCGCA
Human	GACAATTCTGCTAGCAGCCTTTGTGCTATTATCTGTTTCTAAACTTAGTAATTGAGTGT
Chimpanzee	GACAATTCTGCTAGCAGCCTTTGTGCTATTATCTGTTTCTAAACTTAGTAATTGAGTGT
Macaque	GACAATTCTGCTAGCAGCCTTTGTGCTATTATCTGTTTCTAAACTTAGTAATTGAGTGT
Human	GATCTGGAGACTAA-CTCTGAAATAAATAAGCTGATTATTTATTTATTTCTCAAAACAA
Chimpanzee	GATCTGGAGACTAACTCTGAAATAAATAAGCTGATTATTTATTTATTTCTCAAAACAA
Macaque	TATCTGGAGACTAACTCTGAAATAAATAAGCTGATTATTTATTTATTTCTCAAAACAA
Human	CAGAATACGATTAGCAAAATAGTCTTAAAGATATTTATTTACATTTCTATATTTCTCTA
Chimpanzee	CAGAATACGATTAGCAAAATAGTCTTAAAGATATTTATTTACATTTCTATATTTCTCTA
Macaque	CAGAATATGATTAGCAAAATAGTCTTAAAGATATTTATTTGACTTCTATATTTCTCTA
Human	CCCTGAGTTGATGTGTGAGCAATATGTCACCTTTCATAAAGCCAGGTATACA...TTATG
Chimpanzee	CCCTGAGTTGATGTGTGAGCGTATGTCACCTTTCATAAAGCCAGGTATACA...TTATG
Macaque	CCCTGAGTTGATGTGTGAGCAATATGTCACCTTTCATAAAGCCAGGTATATACATTACG
Human	GACAGGTAAGTAAAAACATATTATTTATCTACGTTTTTGTCCAAAATTTTAAATTTTC
Chimpanzee	GACAGGTAAGTAAAAACATATTATTTATCTACGTTTTTGTCCAAAATTTTAAATTTTC
Macaque	GACAGGTAAGTAAAAA-CATATTATTTATCTAGTTTTTGTCCAAAGATTTTAAATTTTC
Human	AACTGTTGCGGCTGTGTTGGTAA...TGTAAAAACAATCTAGTACA
Chimpanzee	AACTGTTGCGGCTGTGTTGGTAA...TGTAAAAACAATCTAGTACA
Macaque	AACTGTTGTTGATGTTGTTGGTAA...CTAAAAACAATCTAGTACG

Types of Data Sets: (4) Spatial, image and multimedia Data

- **Spatial data:** . An example of spatial data is weather data (precipitation, temperature, pressure) that is collected for a variety of geographical locations.

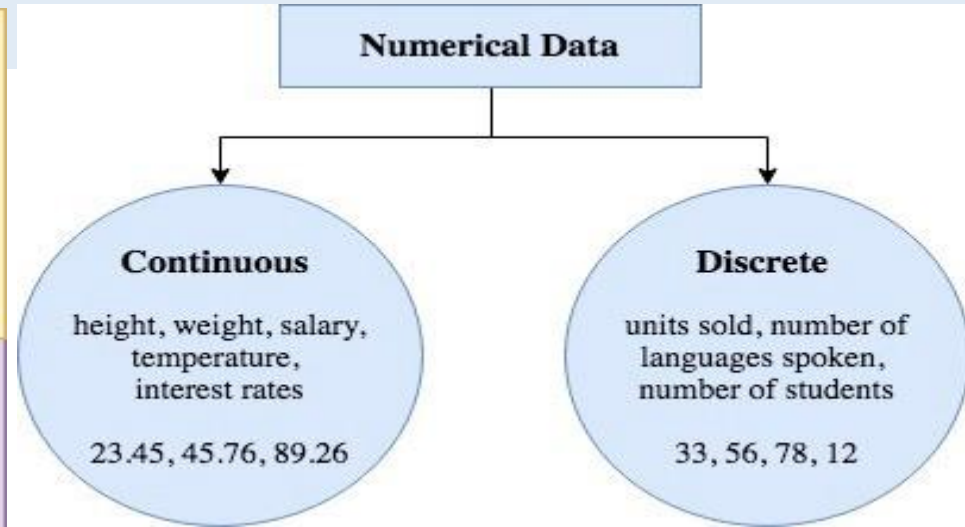


- **Image data:**
- **Video data:**



Data Types From A Machine Learning Perspective

Numerical Data	Categorical Data
Time Series Data	Text



Continuous data can assume any value within a range whereas discrete data has distinct values

Numerical Data

- Data points are exact numbers
- Quantitative data
- Measurement - number of residential properties in Los Angeles or how many houses sold in the past year.

Numerical data is not ordered in time

- Number of students taking machine learning class would be a discrete data set.
- You can only have discrete whole number values like 10, 25, or 33.
- A class cannot have 12.75 students enrolled, A student either join a class or he doesn't

Continuous data are numbers that can fall anywhere within a range.

Like a student could have an average score of 88.25 which falls between 0 and 100

Data Types From A Machine Learning Perspective

Numerical Data	Categorical Data
Time Series Data	Text

- **Nominal categorical:** No concept of ordering amongst the values of that attribute

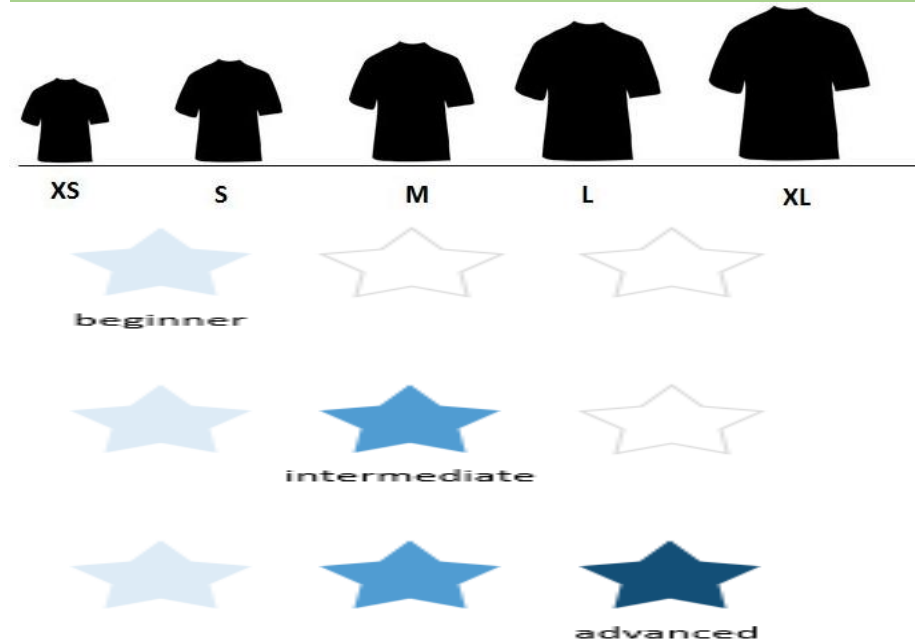
sunny 	cloudy 	snowy 
rainy 	windy 	icy 

Similarly movie, music and video game genres, country names, food and cuisine types are other examples of nominal categorical attributes

- **Ordinal categorical :** Sense or notion of order amongst its values

Categorical Data

- Represents characteristics – cricket player's position, team, hometown
- Belong to a specific finite set of categories or classes -- classes or labels
- Major classes of categorical data => nominal and ordinal



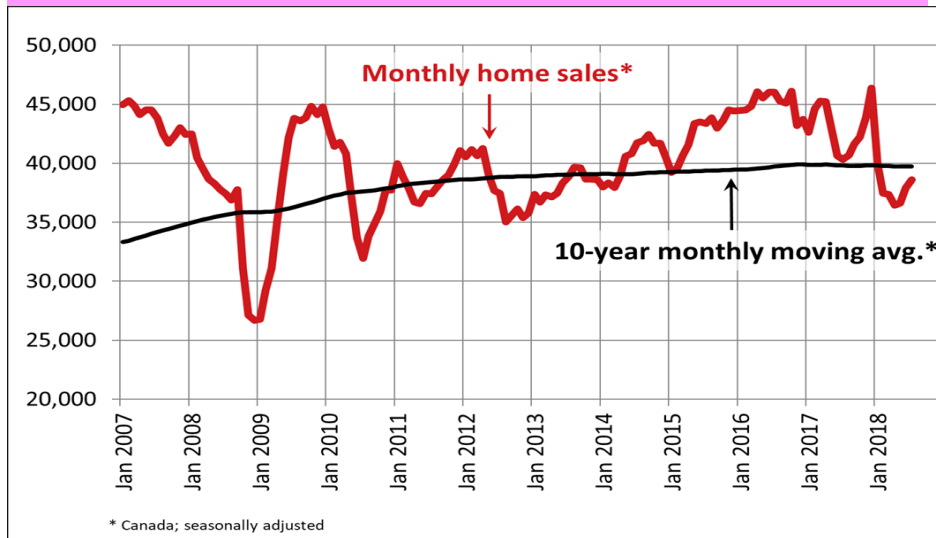
Data Types From A Machine Learning Perspective

Numerical Data	Categorical Data
Time Series Data	Text

- For example
 - Measure the average number of home sales for many years.
- The difference of time series data and numerical data is that rather than having a bunch of numerical values which don't have any time ordering, time series data does have some implied ordering.
- There is a first data point collected and a last data point collected.

Time Series Data

- Sequence of numbers collected at regular intervals over some period of time
- Time series data has a temporal value attached to it, so this would be something like a date or a time stamp that you can look for trends in time



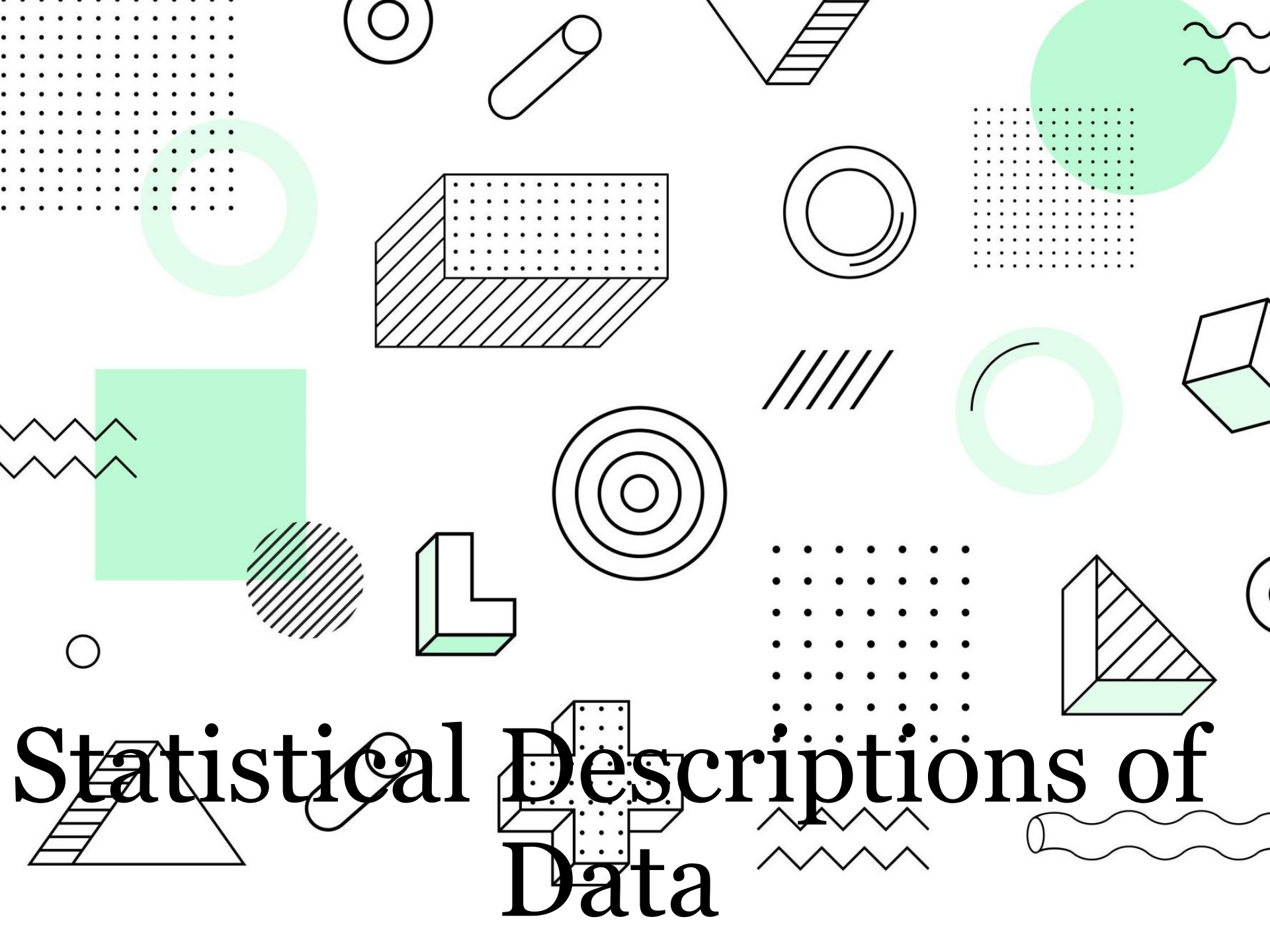
Data Types From A Machine Learning Perspective

Numerical Data	Categorical Data
Time Series Data	Text

Text

- Text data is basically just words.
- A lot of the time the first thing that you do with text is you turn it into numbers using some interesting functions like the bag of words formulation.





Statistical Descriptions of Data

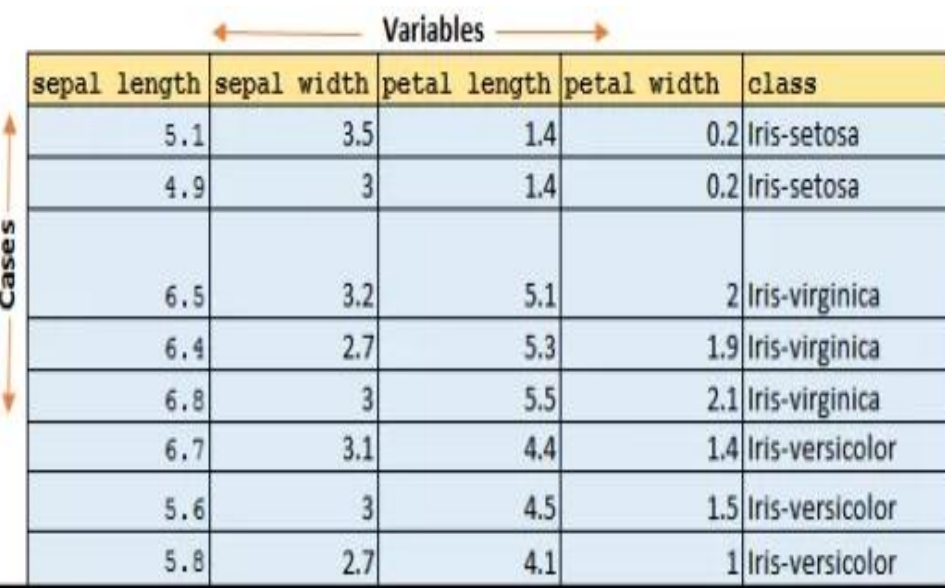
Basic Statistical Descriptions of Data

Data Matrix- tabular format representation of cases and variables

- Each row of a data matrix represents a case and each column represent a variable
- A complete Data Matrix may contain thousands or lakhs or even more cases.

Frequency table

- Shows how the values of a variable are distributed over the cases.



sepal length	sepal width	petal length	petal width	class
5.1	3.5	1.4	0.2	Iris-setosa
4.9	3	1.4	0.2	Iris-setosa
6.5	3.2	5.1	2	Iris-virginica
6.4	2.7	5.3	1.9	Iris-virginica
6.8	3	5.5	2.1	Iris-virginica
6.7	3.1	4.4	1.4	Iris-versicolor
5.6	3	4.5	1.5	Iris-versicolor
5.8	2.7	4.1	1	Iris-versicolor

Class	Frequency	Percentage	Cumulative Percentage
Iris-setosa	2	25%	25%
Iris-virginica	3	38%	63%
Iris-versicolor	3	38%	100%
Total	8	100%	

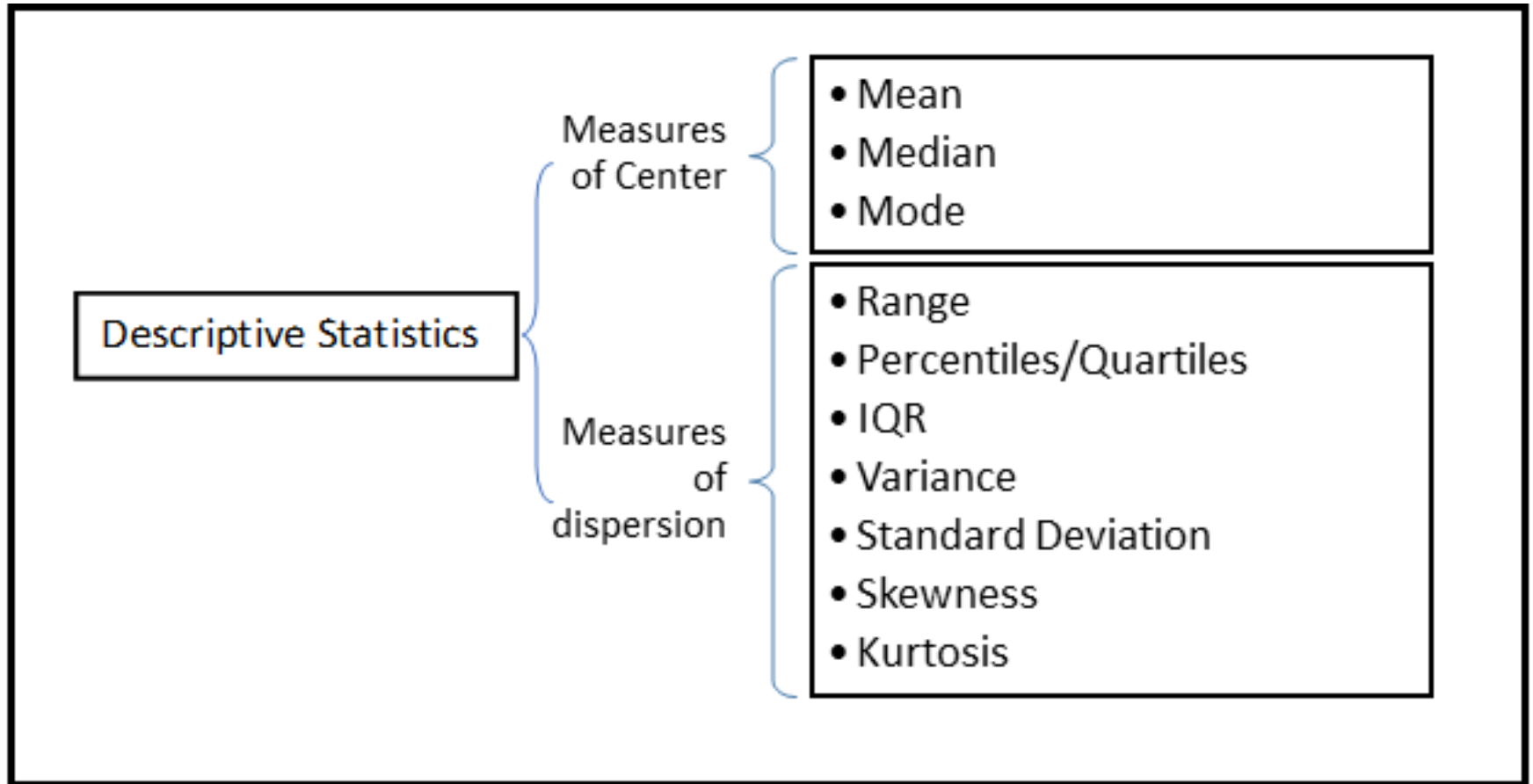
Types of statistics

- Descriptive
 - Describe the basic features of data- simple summaries
 - Data Analysis
 - Analyze the previous year sales data to find interesting insight
 - Invest made by the company
 - Profit % etc
 - Use numerical measures
- Inferential
 - Get inferences and predictions about the larger population from which the sample was drawn
 - Data Science
 - Use data to find several insights/inferences
 - Suggest the company about certain strategies to increase their profit

Basic Statistical Descriptions of Data

- What
 - Measure of central tendency
 - Mean, Median and mode
 - Location of the centre of a data distribution
 - Where do **most of the attribute values fall?**
 - Dispersion Measure
 - Range, quartiles, inter quartile range, five number summary and box plots , variance and standard deviation
 - It describes **how are the data spread out.**

Descriptive Statistics



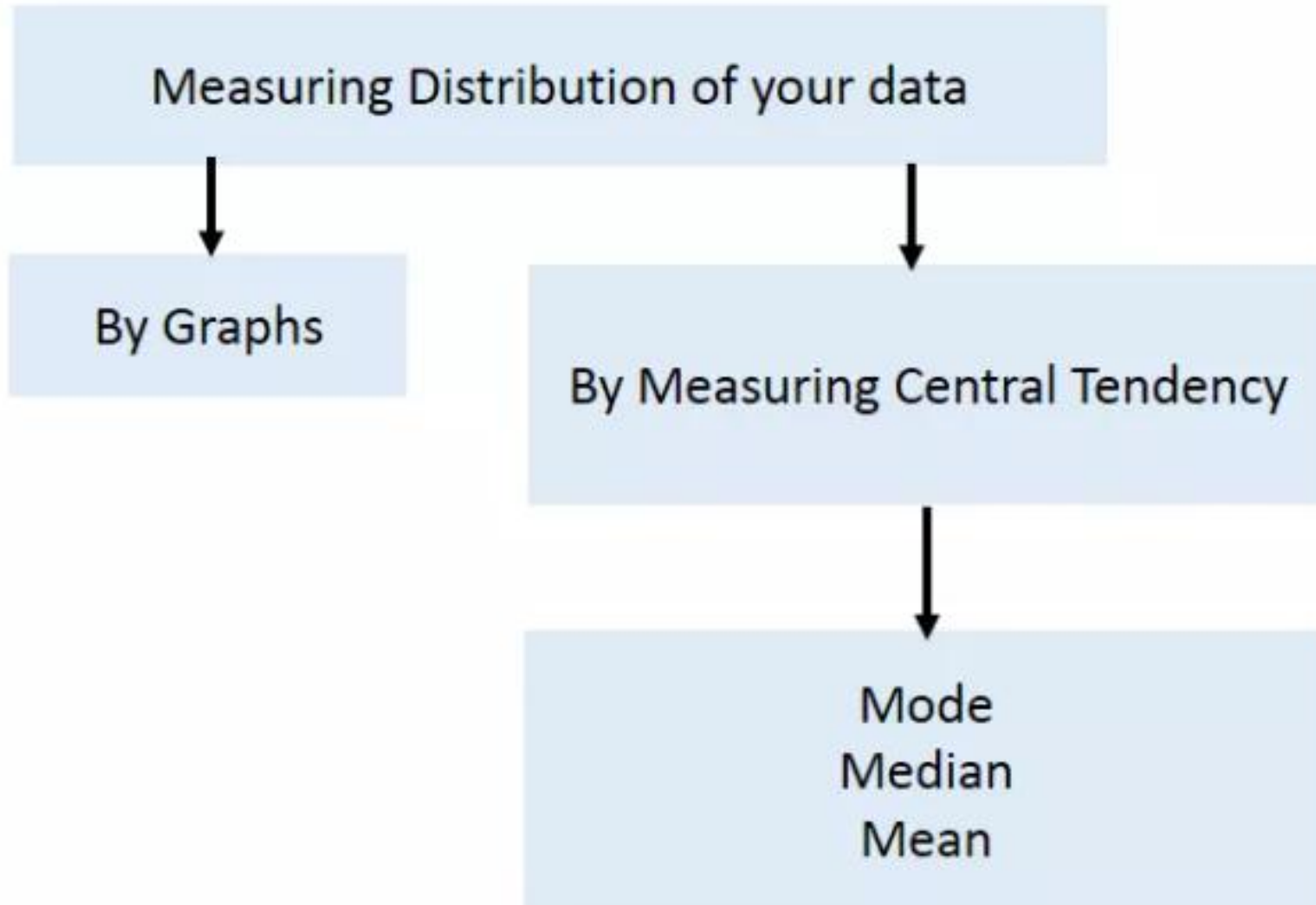
Measuring Central Tendency

- What single number best represents our data?
- Whether the data are packed together?

Central tendency : indicates where the centre of the distribution tends to be. eg., Measure of scores, Measure of height

- Measure of central tendency answers whether the scores are high or low.
- Determining which measure of central tendency to use depends on:
 - Scale of measurement (NOIR)
 - Shape of the distribution (skew, kurtosis)

Measuring Distribution of Your Data



Measuring Distribution of Your Data

Mode

- Value that occurs most frequently in the data
- Unimodal, bimodal, multimodal

5, 6, 5, 7, 5, 8, 9, 5

Mode

Mode is 5 here.
Because It is most frequent item.

5, 6, 5, 6, 5, 8, 6, 5, 6

Mode

Here 5 and 6 both are mode. Because both are most frequent items.

Contd...

The Mode : The most frequently occurring score

- Unimodal distribution has only **one** major peak
- Bimodal distribution has **two** major peaks
- **Mode commonly used with nominal data**

Dataset A : 2 3 3 4 4 4 4 7 7 8 9 : 4

Dataset B : 2 3 4 4 4 4 7 8 9 9 9 9 10 12 : 4 and a warning

- Not susceptible to **outliers** at all

Outlier : significantly higher or lower than other scores

- Mode ignores much of the data
- Used in non-parametric methods.

The Median

Median : Geographic centre of the dataset

- If the median is 30, then half the scores are at or below 30
- Put the data in ascending order
- Find the middle number
 - If n is odd, the median located at $(n + 1)/2$ position
 - If n is even, the median will be average of two middle numbers. $\text{Avg}(n/2, n/2 + 1)$

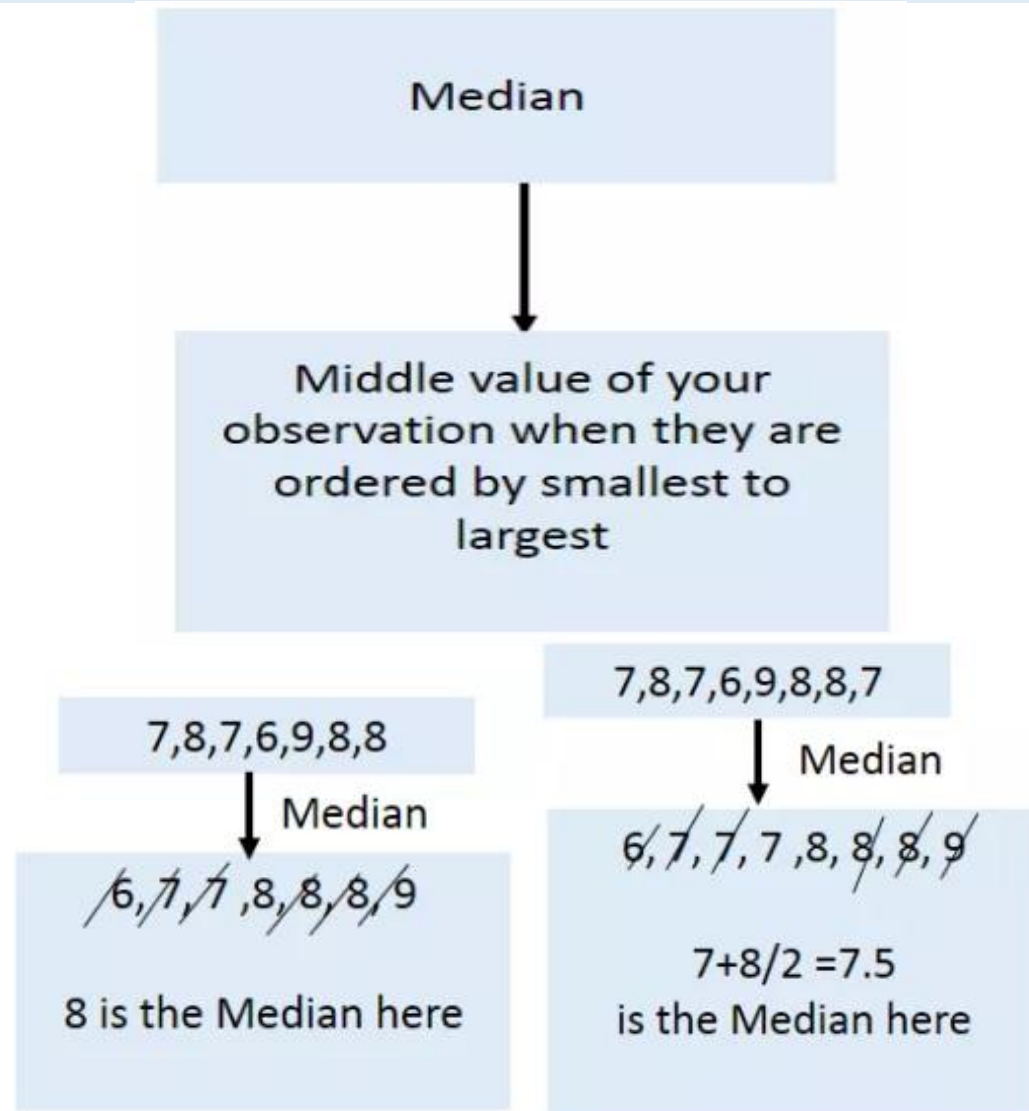
Contd...

Median

- Median is nothing more than the middle value of your observations when they are order from the smallest to the largest.

It involves two steps:

- Oder your cases from smallest to largest
- Find the middle Value



The Mean

- Mean is located at the mathematical centre of the distribution.
- Most commonly used with Interval and Ratio data
- It is also called average
- In normal distribution Mean = Median = Mode

$$\text{Mean} = M = \frac{\Sigma(X)}{N} = \bar{X}$$

Contd...

- Mean- The mean is the sum of all the values divided by the number of observations

7,8,7,6,9,8,8



Mean

$$\text{Sum}(53) / \text{\#cases } (7) \\ = 7.6$$

When to use what
measurement of central
tendency ??

- If data is Categorical (Nominal or Ordinal) it is impossible to calculate mean or median. So, go for **mode**.
- If your data is quantitative then go for mean or median.
- Basically, if your data is having some influential outliers or data is highly skewed then **median** is the best measurement for finding central tendency. Otherwise go for **Mean**

Whether the data are packed
together or spread-out?

Dispersion or Variability of Data

Variability indicated how spread out the scores are

- When there are large differences in the scores, the data are said to contain a lot of Variability.

An example:

Consider you are driving cross district and eating every A2B restaurant that you pass. Do you expect the variability of the food taste from one restaurant to the next will be high or low?



Dispersion or Variability of Data

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An example:

Consider you are driving cross district and eating every A2B restaurant that you pass. Do you expect the variability of the food taste from one restaurant to the next will be high or low?

Variability will be low and consistency will be high. The food tastes similar.

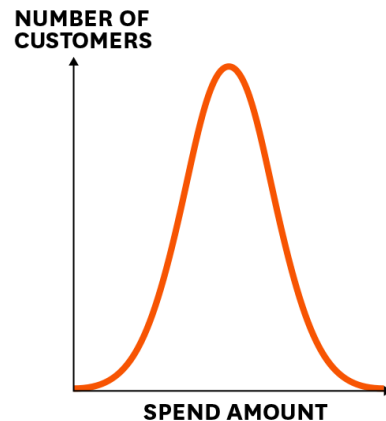


Dispersion or Variability of Data

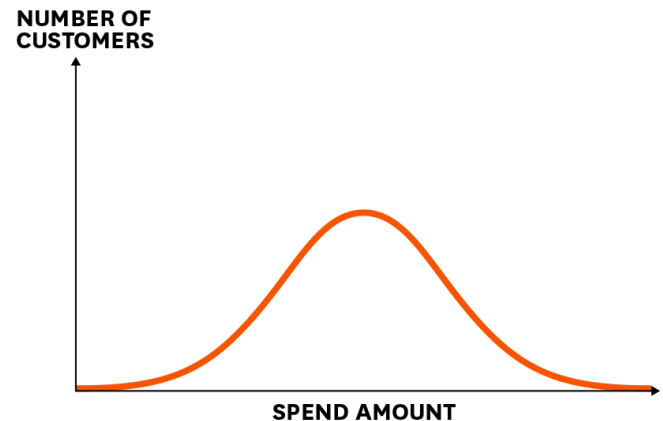
The **grater the variability**, the less accurately the data are summarized by the measures of **central tendency**

Measurement errors are larger, the mean is not useful

Lesser Variation



Greater Variation



Dispersion Vs Central Tendency

Dispersion and Central tendency are independent measures

Mean tells where the centre is

- Distribution with same mean may have different variability

0, 2, 6, 10, 12 : Mean = 6 (High variability)

8, 7, 6, 5, 4 : Mean = 6 (Data points are closer)

6, 6, 6, 6, 6 : Mean = 6 (No variability)

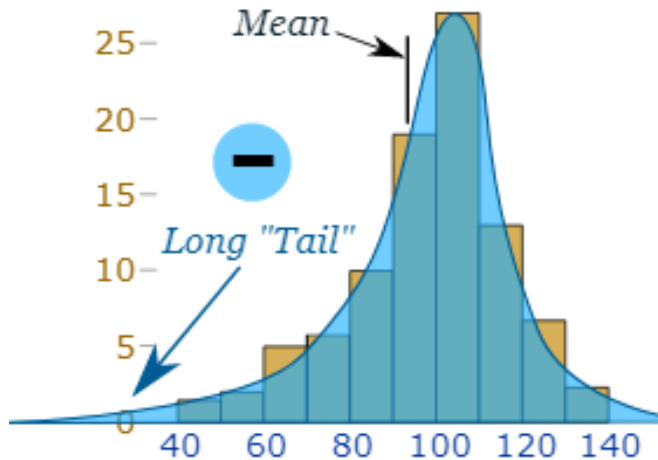
Measures of dispersion - Skewness

- To check whether the distribution has a longer tail on one side or the other or has left-right symmetry.
- It can be positive(representing **right skewed** distribution), negative(representing **left skewed** distribution), or zero(representing **unskewed** distribution).

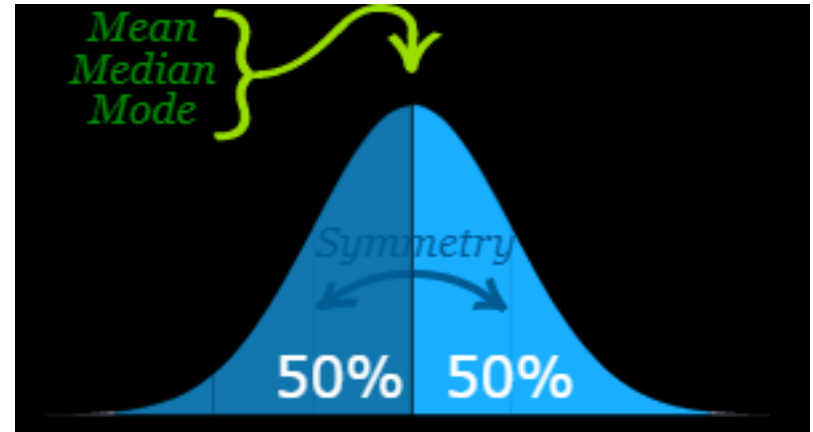
Symmetric vs. Skewed Data

- Data can be "skewed", meaning it tends to have a **long tail** on one side or the other:

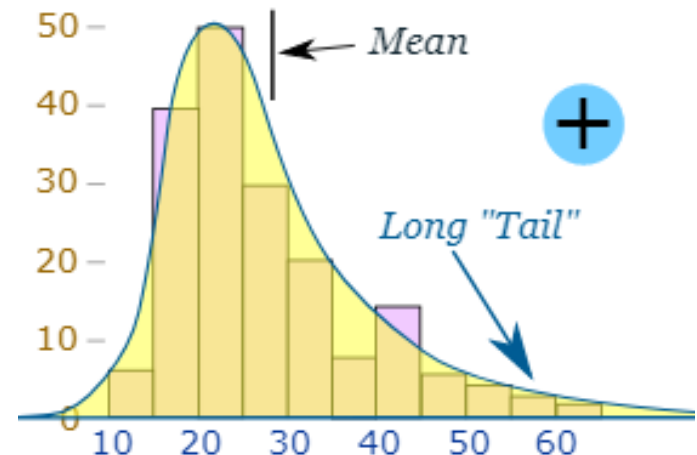
Negative Skew



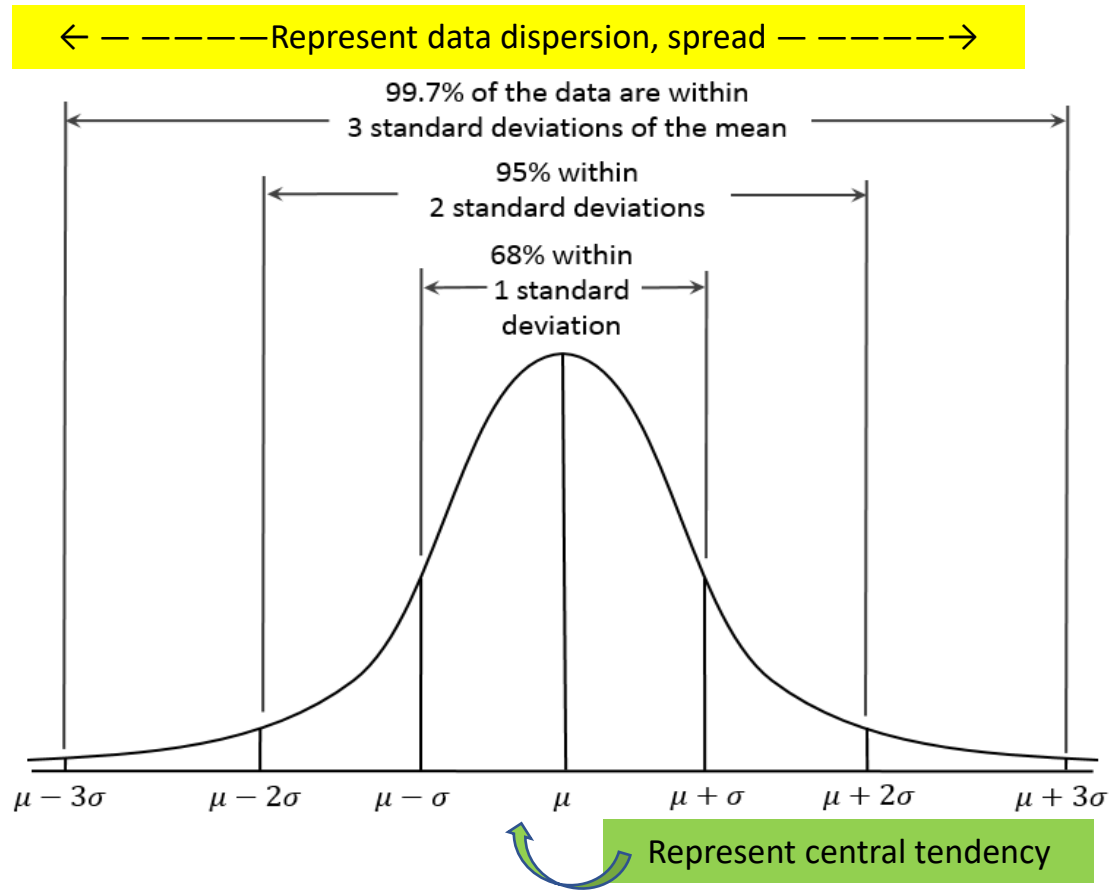
No Skew / symmetric



Positive Skew

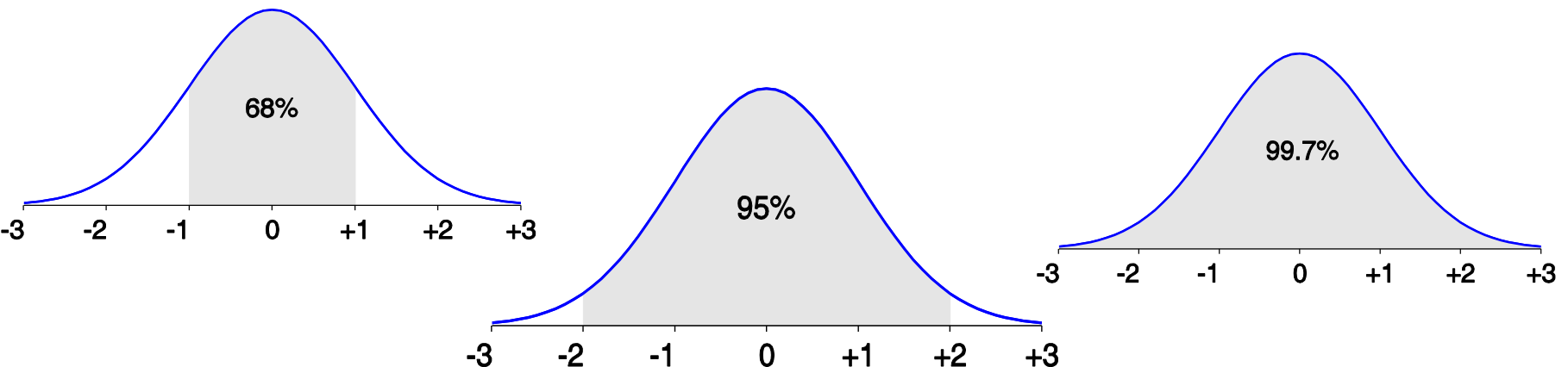


Properties of Normal Distribution Curve



Properties of Normal Distribution Curve

- The normal (distribution) curve
 - From $\mu - \sigma$ to $\mu + \sigma$: contains about 68% of the measurements (μ : mean, σ : standard deviation)
 - From $\mu - 2\sigma$ to $\mu + 2\sigma$: contains about 95% of it
 - From $\mu - 3\sigma$ to $\mu + 3\sigma$: contains about 99.7% of it



Measuring the Dispersion of Data

Range

- The difference between the largest and the smallest data item.

Let's look at a very simple set of data representing the weight of 10 males,

55, 56, 56, 58, 60, 61, 63, 64, 70, 78.

$$\text{Range} = 78 - 55 = 23.$$

Standard deviation

Square root of the variance. In the above formula, σ is the standard deviation and σ^2 is the variance
Std dev = $\sqrt{51.88} = 7.20$

Variance

- Variance signifies how much the data items are deviating from mean
 - Larger variance means the data items deviate more from the mean.
 - Smaller variance means the data items are closer to the mean.

$$\text{Variance}(\sigma^2) = \sum \frac{(x - \bar{x})^2}{n - 1}$$

$$\begin{aligned} &= [(55 - 62.1)^2 + (56 - 62.1)^2 + (56 - 62.1)^2 \\ &\quad + (58 - 62.1)^2 + (60 - 62.1)^2 + (61 - 62.1)^2 \\ &\quad + (63 - 62.1)^2 + (64 - 62.1)^2 + (70 - 62.1)^2 \\ &\quad + (78 - 62.1)^2] / 9 \\ &= 466.9 / 9 \\ &= 51.88. \end{aligned}$$

Measuring the Dispersion of Data

Correlation - Co-variance

- Measures how two variables vary with respect to each other

$$\text{Covar}(X, Y) = \sum \frac{(x - \bar{x})(y - \bar{y})}{n - 1}$$

Covariance indicates the direction of the linear relationship between variables. ($-\infty$ and $+\infty$)

Covariance of two dependent variables measures how much in real quantity (i.e. cm, kg, liters) on average they co-vary.

Positive covariance signifies that the higher values of one variable correspond with the higher values of the other variable, and similarly for the lower ones.

Negative covariance, on the other hand, signifies that the higher values of one variable correspond to the lower values of the other

Measuring the Dispersion of Data

	S&P 500	ABC Corp.
2013	1,692	68
2014	1,978	102
2015	1,884	110
2016	2,151	112
2017	2,519	154

$$\text{Mean (S\&P 500)} = \frac{1,692 + 1,978 + 1,884 + 2,151 + 2,519}{5} = 2,044.80$$

$$\text{Mean (ABC Corp.)} = \frac{68 + 102 + 110 + 112 + 154}{5} = 109.20$$

	S&P 500	ABC Corp.	a	b	a x b
2013	1,692	68	-352.80	-41.20	14,535.36
2014	1,978	102	-66.80	-7.20	480.96
2015	1,884	110	-160.80	0.80	-128.64
2016	2,151	112	106.20	2.80	297.36
2017	2,519	154	474.20	44.80	21,244.16
Mean	2,044.80	109.20	Sum		36,429.20

$$\text{Cov(S\&P 500, ABC Corp.)} = \frac{36,429.20}{5 - 1} = 9,107.30$$

Positive covariance and as such it means that both returns move in the same direction
i.e. either both have positive returns or both have negative returns

A covariance very close to zero signifies the lack of correlation between two variables.

Measuring the Dispersion of Data

Correlation - how two variables move with respect to each other

Correlation on the other hand measures both the strength and direction of the linear relationship between two variables. (-1 to 1)

Correlation of two dependent variables measures the proportion of how much on average these variables vary w.r.t one another.

A perfect positive correlation means that the correlation coefficient is 1.

A perfect negative correlation means that the correlation coefficient is -1.

A correlation coefficient of 0 means that the two variables are independent of each other

• The correlation coefficient

$$r_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x}) \sum_{i=1}^n (y_i - \bar{y})}}$$

$$\text{Correlation}(X, Y) = \frac{\text{Covar}(X, Y)}{\sigma_x \sigma_y} = \sum \frac{(x - \bar{x})(y - \bar{y})}{\sigma_x \sigma_y (n - 1)}$$

Measuring the Dispersion of Data

2 *Subtract Mean*

3 *Calculate ab , a^2 and b^2*

Temp °C	Sales	"a"	"b"	a×b	a ²	b ²
14.2	\$215	-4.5	-\$187	842	20.3	34,969
16.4	\$325	-2.3	-\$77	177	5.3	5,929
11.9	\$185	-6.8	-\$217	1,476	46.2	47,089
15.2	\$332	-3.5	-\$70	245	12.3	4,900
18.5	\$406	-0.2	\$4	-1	0.0	16
22.1	\$522	3.4	\$120	408	11.6	14,400
19.4	\$412	0.7	\$10	7	0.5	100
25.1	\$614	6.4	\$212	1,357	41.0	44,944
23.4	\$544	4.7	\$142	667	22.1	20,164
18.1	\$421	-0.6	\$19	-11	0.4	361
22.6	\$445	3.9	\$43	168	15.2	1,849
17.2	\$408	-1.5	\$6	-9	2.3	36
18.7	\$402			5,325	177.0	174,757

1 *Calculate Means*

4 *Sum Up*

5
$$\frac{5,325}{\sqrt{177.0 \times 174,757}} = 0.9575$$

Measures of dispersion - Percentile

- A measure which indicates the value below which a given percentage of points in a dataset fall.
- The 35th percentile(P_{35}) is the score below which 35% of the data points may be found.
- Median represents the 50th percentile, 0th percentile representing the minimum and 100th percentile representing the maximum of all data points.

To calculate kth percentile(P_k) for a data set of N observations which is arranged in increasing order,

Step 1: Calculate $i = (k/100) \times N$

Step 2: If i is a whole number, then count the observations in the data set from left to right till we reach the i^{th} data point. The k^{th} percentile, in this case, is equal to the average of the value of i^{th} data point and the value of the data point that follows it

Step 3: If i is not a whole number, then round it up to the nearest integer and count the observations in the data set from left to right till we reach the i^{th} data point. The k^{th} percentile now is just equal to the value corresponding this data point.

Measures of dispersion - Percentile Example

- There are 25 test scores such as: 72, 54, 56, 61, 62, 66, 68, 43, 69, 69, 70, 71, 77, 78, 79, 85, 87, 88, 89, 93, 95, 96, 98, 99, 99. Find the 60th percentile?
- **Step 1:** Arrange the data in the ascending order. **Ascending Order** = 43, 54, 56, 61, 62, 66, 68, 69, 69, 70, 71, 72, 77, 78, 79, 85, 87, 88, 89, 93, 95, 96, 98, 99, 99.
- **Step 2:** Find Rank, $\text{Rank} = \text{Percentile} / 100 = 60 / 100$ $k = 0.60$
- **Step 3:** Find 60th percentile, $60\text{th percentile} = 0.60 \times 25 = 15$
- **Step 4:** Count the values in the given data set from left to right until you reach the number 15. From the given data set, 15th number is 79. Now take the 15th number and the 16th number and find the average: $79 + 85 / 2 = 164 / 2 = 82$
- **Hence, 60th percentile of given data set = 82.**

Measures of dispersion - Percentile

Let us consider the percentile example problem: In a college, a list of grades of 15 students has been declared. Their grades are: 85, 34, 42, 51, 84, 86, 78, 85, 87, 69, 74, 65. Find the 80th percentile?

- **Step 1:** Arrange the data in the ascending order.
- **Ascending Order** = 34, 42, 51, 65, 69, 74, 78, 84, 85, 85, 86, 87.
- **Step 2:** Find Rank, $\text{Rank} = \text{Percentile} / 100 = 80 / 100$ $k = 0.80$
- **Step 3:** Find 80th percentile, $80\text{th percentile} = 0.80 \times 12 = 9.6$
- **Step 4:** Since it is not a whole number, round to the nearest whole number. Therefore, 9.6 is rounded to 10.
- Now, count the values in the given data set from left to right until you reach the number 10. From the given data set, 10th number is 85.
Hence,
- **80th percentile of given data set = 85**

Measures of dispersion - Quartiles

- Three points that split the data set into four equal parts such that each group consists of one-fourth of the data.

25th percentile the first quartile(Q_1), 50th percentile the second quartile(Q_2), and 75th percentile the third quartile(Q_3).

Why percentiles?

Percentile gives the relative position of a particular value within the dataset. If we are interested in relative positions, then mean and standard deviations does not make sense. In the case of exam scores, we do not know if it might have been a difficult exam and 7 points out of 20 was an amazing score. In this case, personal scores in itself are meaningless, but the percentile would reflect everything. For example, GRE and GMAT scores are all in terms of percentiles.

Characteristic of Range

- Rarely used
- Its **crude** measure
- Highly susceptible to **outliers**
- Used mostly with nominal data or ordinal data
- The Interquartile Range (IQR) overcome these limitations to some extent

Measures of dispersion - Interquartile Range(IQR)

- The difference between the third quartile and the first quartile.

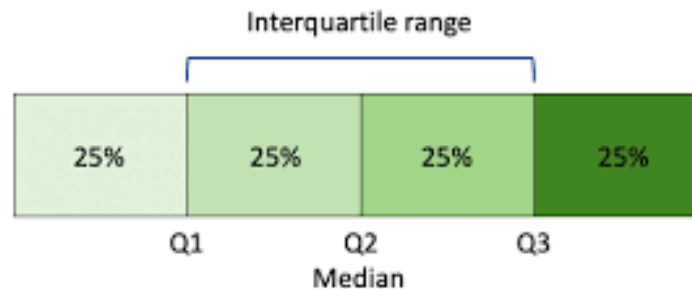
$$\text{IQR} = Q_3 - Q_1$$

Why IQR?

The interquartile range is a better option than range because it is not affected by outliers. It removes the outliers by just focusing on the distance within the middle 50% of the data

Interquartile Range

- Quartiles divide the data into four equal sections (25% each)
- Interquartile Range - The range between middle 50% of data
 - Measure the variability between 1st and 3rd quartile
 - Variability in the middle half of the data
 - Describes spread at centre of the data
 - Not largely affected by the outliers



Calculating Interquartile Range

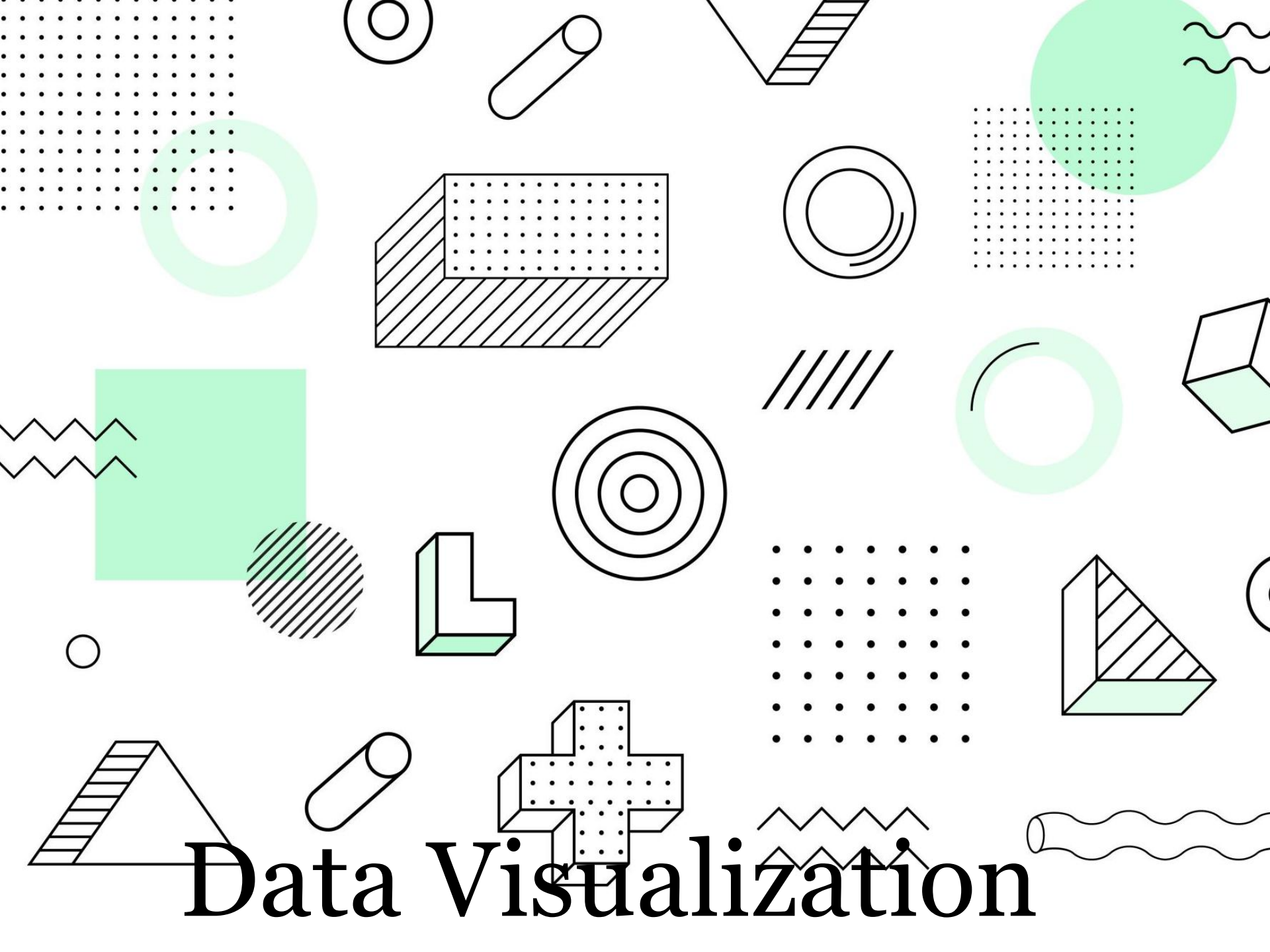
- Median split 1 and 2
- Median split both lower and upper median
- subtract lower from upper

Data Set D:



$$\text{MDN} = (3 + 4) / 2 = 3.5$$

$$\text{IQR} = 9 - 3 = 6$$



Data Visualization

Graphic Displays of Basic Statistical Descriptions

- **Boxplot:** graphic display of five-number summary
- **Histogram:** x-axis are values, y-axis repres. frequencies
- **Scatter plot:** each pair of values is a pair of coordinates and plotted as points in the plane

The 5 Number Summary

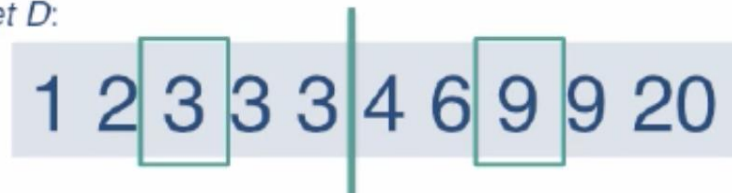
- Five numbers summaries the dataset

Minimum, Quartile1 (Q1), Median, Quartile3 (Q3),
Maximum

Q1 : First quartile, 25th percentile, middle score
between minimum and median

Q2 : Third quartile, 75th percentile, middle of scored
between Median and Maximum

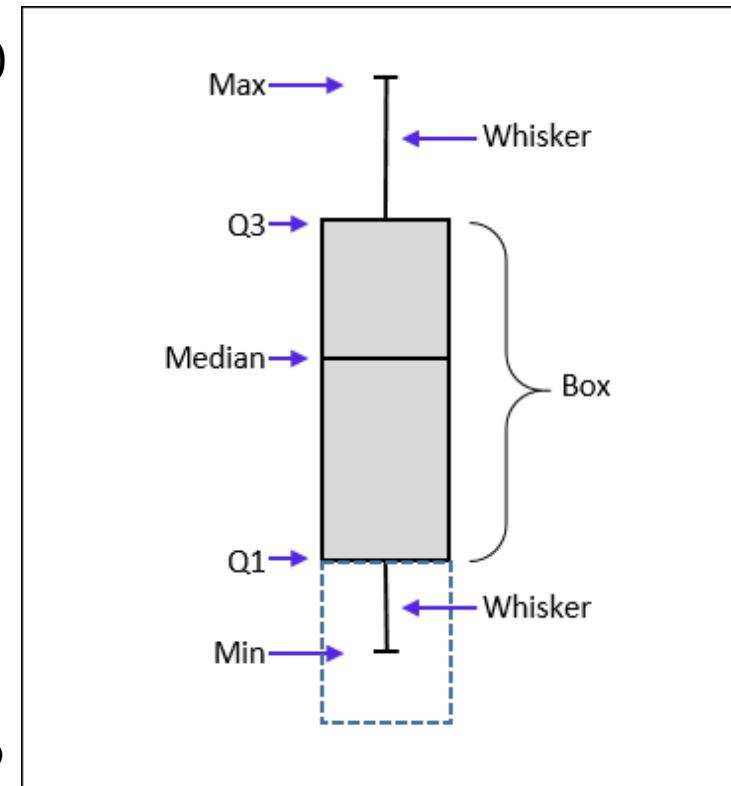
Data Set D:



Min = 1, Q1 = 3, MDN = 3.5, Q2 = 9, max = 20

Measuring the Dispersion of Data: Quartiles & Boxplots

- **Quartiles:** Q_1 (25th percentile), Q_3 (75th percentile)
- **Inter-quartile range:** $IQR = Q_3 - Q_1$
- **Five number summary:** min, Q_1 , median, Q_3 , max
- **Boxplot:** Data is represented with a box
 - Q_1 , Q_3 , IQR: The ends of the box are at the first and third quartiles, i.e., the height of the box is IQR
 - Median (Q_2) is marked by a line within the box
 - Whiskers: two lines outside the box extended to Minimum and Maximum



- ❑ Outliers: points beyond a specified outlier threshold, plotted individually
- ❑ **Outlier: usually, a value higher/lower than $1.5 \times IQR$**

Measuring the Dispersion of Data: Quartiles & Boxplots

A sample of 10 boxes of raisins has these weights (in grams):

25, 28, 29, 29, 30, 34, 35, 35, 37, 38

Make a box plot of the data.

Step 1: Order the data from smallest to largest.

Our data is already in order. 25, 28, 29, 29, 30, 34, 35, 35, 37, 38

Step 2

Find the median.

The median is the mean of the middle two numbers:

$$30 + 34 / 2 = 32 \quad \text{median} = 32$$

Step 3: Find the quartiles.

The **first quartile** is the median of the data points to the left of the median.

$$Q_1 = 29$$

The **third quartile** is the median of the data points to the right of the median.

$$Q_3 = 35$$

Step 4: Complete the five-number summary by finding the min and the max.

The min is the smallest data point, which is 25.

The max is the largest data point, which is 38.

The five-number summary is 25, 29, 32, 35, 38.

Constructing a box and whisker plot :

Example 2

- Step 1 - take the set of numbers given...**34, 18, 100, 27, 54, 52, 93, 59, 61, 87, 68, 85, 78, 82, 91**

Place the numbers in order from least to greatest:**18, 27, 34, 52, 54, 59, 61, 68, 78, 82, 85, 87, 91, 93, 100**

- Step 2 - Find the median. Remember, the median is the middle value in a data set.
18, 27, 34, 52, 54, 59, 61, 68, 78, 82, 85, 87, 91, 93, 100

68 is the median of this data set.

- Step 3 – Find the lower quartile. The lower quartile is the median of the data set to the left of 68. **(18, 27, 34, 52, 54, 59, 61,) 68, 78, 82, 85, 87, 91, 93, 100**

52 is the lower quartile

- Step 4 – Find the upper quartile. The upper quartile is the median of the data set to the right of 68. **18, 27, 34, 52, 54, 59, 61, 68, (78, 82, 85, 87, 91, 93, 100)**

87 is the upper quartile

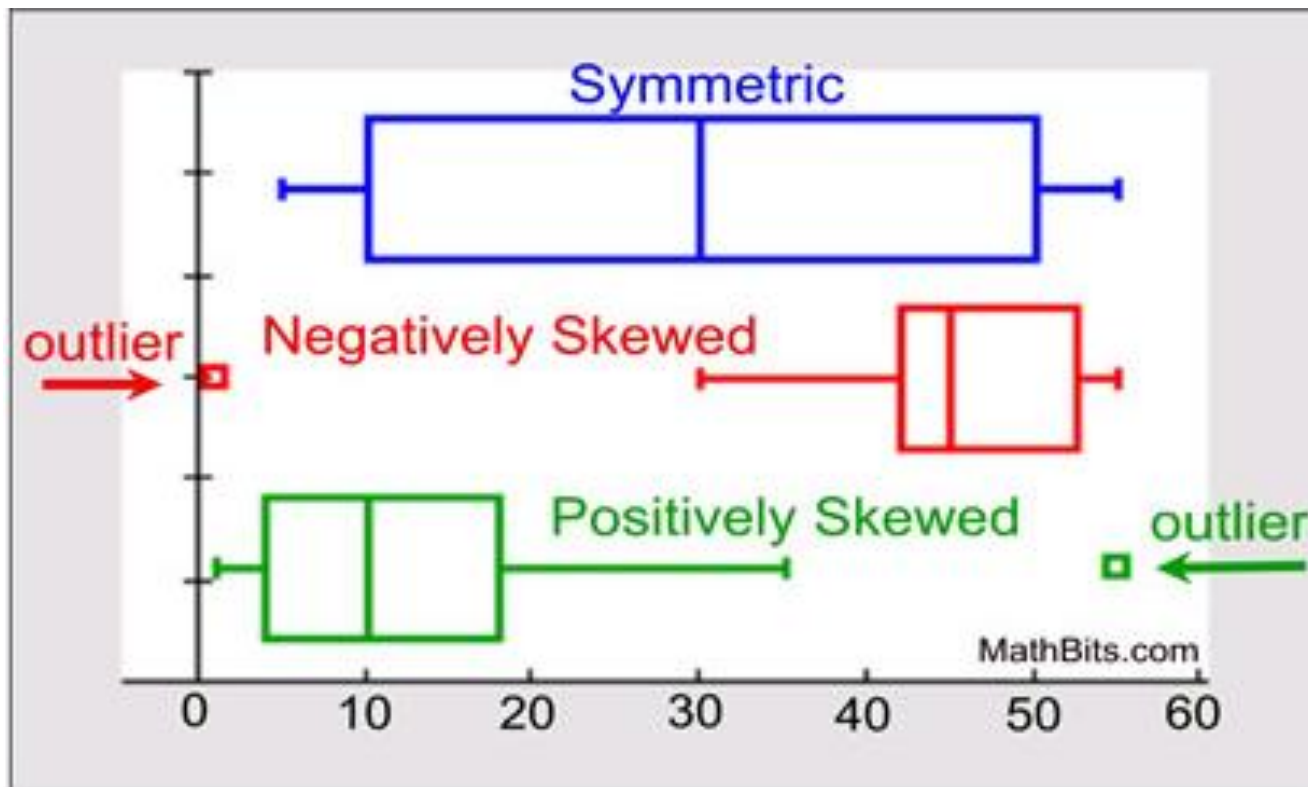
- Step 5 – Find the maximum and minimum values in the set. The maximum is the greatest value in the data set. The minimum is the least value in the data set. **18, 27, 34, 52, 54, 59, 61, 68, 78, 82, 85, 87, 91, 93, 100. 18 is the minimum and 100 is the maximum.**

Constructing a box and whisker plot :

Example 2

- Step 6 – Find the inter-quartile range (IQR).
 - The inter-quartile (IQR) range is the difference between the upper and lower quartiles.
 - Upper Quartile = 87
 - Lower Quartile = 52
 - $87 - 52 = 35$
 - $35 = \text{IQR}$
- Organize the 5 number summary
 - Median – 68
 - Lower Quartile – 52
 - Upper Quartile – 87
 - Max – 100
 - Min – 18

Interpreting the Box Plot:



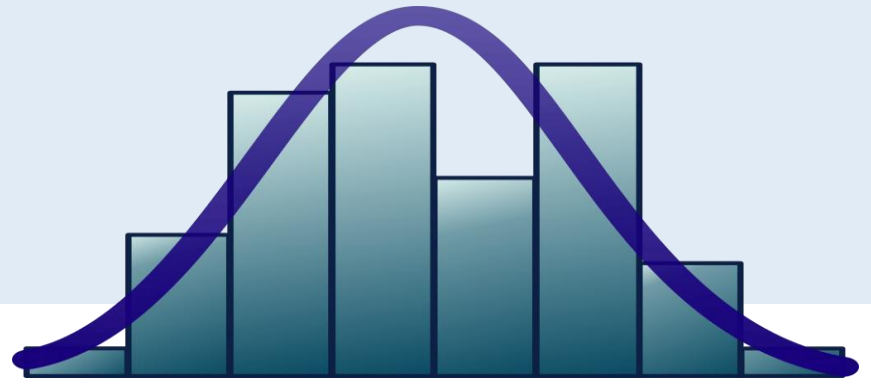
Interpreting the Box Plot:

Symmetric: If a box and whisker plot is symmetric, the **median is equidistant from the minimum and the maximum.**

Negatively Skewed: If a box and whisker plot is negatively skewed, the distance from the **median to the minimum is greater than the distance from the median to the maximum.**

Positively Skewed: If a box and whisker plot is positively skewed, the distance from the **median to the maximum is greater than the distance from the median to the minimum.**

Histogram Analysis



- A histogram is used to summarize discrete or continuous data. In other words, it provides a visual interpretation of numerical data by showing the number of data points that fall within a specified range of values (called “bins”). It is similar to a vertical bar graph. However, a histogram, unlike a vertical bar graph, shows no gaps between the bars.
- Histograms can display a large amount of data and the frequency of the data values. The median and distribution of the data can be determined by a histogram. In addition, it can show any outliers or gaps in the data.

Distributions of a Histogram

- A normal distribution: In a normal distribution, points on one side of the average are as likely to occur as on the other side of the average.

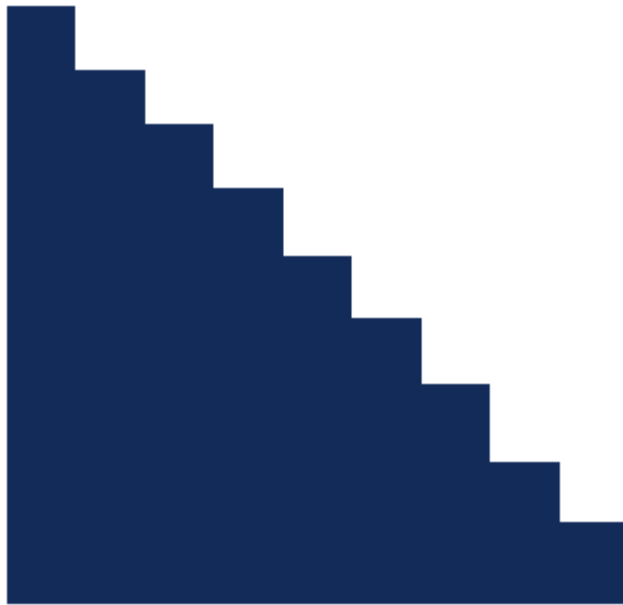


- **A bimodal distribution:** In a bimodal distribution, there are two peaks. In a bimodal distribution, the data should be separated and analyzed as separate normal distributions.



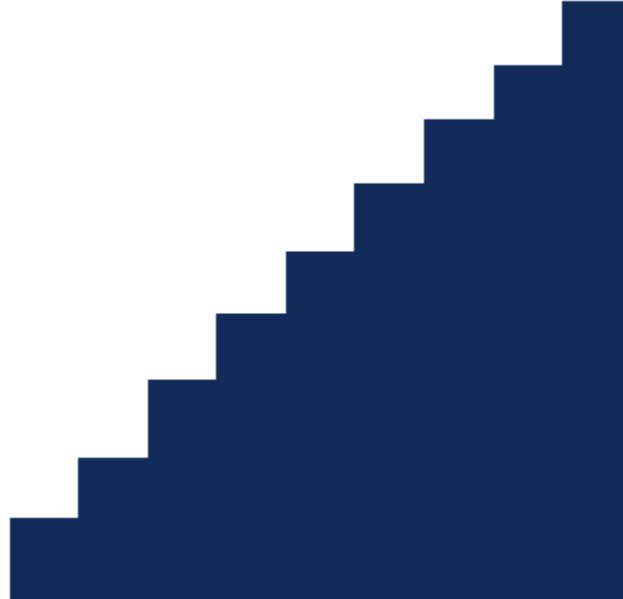
Distributions of a Histogram

- A right-skewed distribution: A right-skewed distribution is also called a positively skewed distribution. In a right-skewed distribution, a large number of data values occur on the left side with a fewer number of data values on the right side. A right-skewed distribution usually occurs when the data has a range boundary on the left-hand side of the histogram. For example, a boundary of 0.



Distributions of a Histogram

- A left-skewed distribution: A left-skewed distribution is also called a negatively skewed distribution. In a left-skewed distribution, a large number of data values occur on the right side with a fewer number of data values on the left side. A right-skewed distribution usually occurs when the data has a range boundary on the right-hand side of the histogram. For example, a boundary such as 100.



Distributions of a Histogram

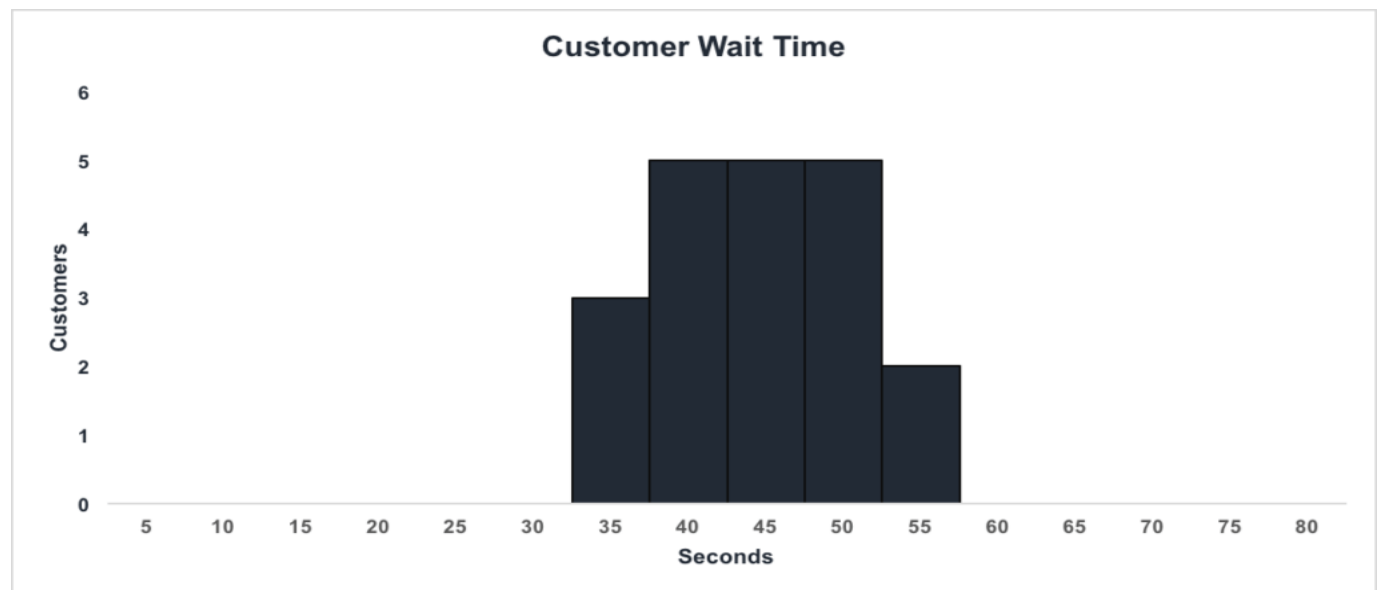
- A random distribution: A random distribution lacks an apparent pattern and has several peaks. In a random distribution histogram, it can be the case that different data properties were combined. Therefore, the data should be separated and analyzed separately.



Example of a Histogram

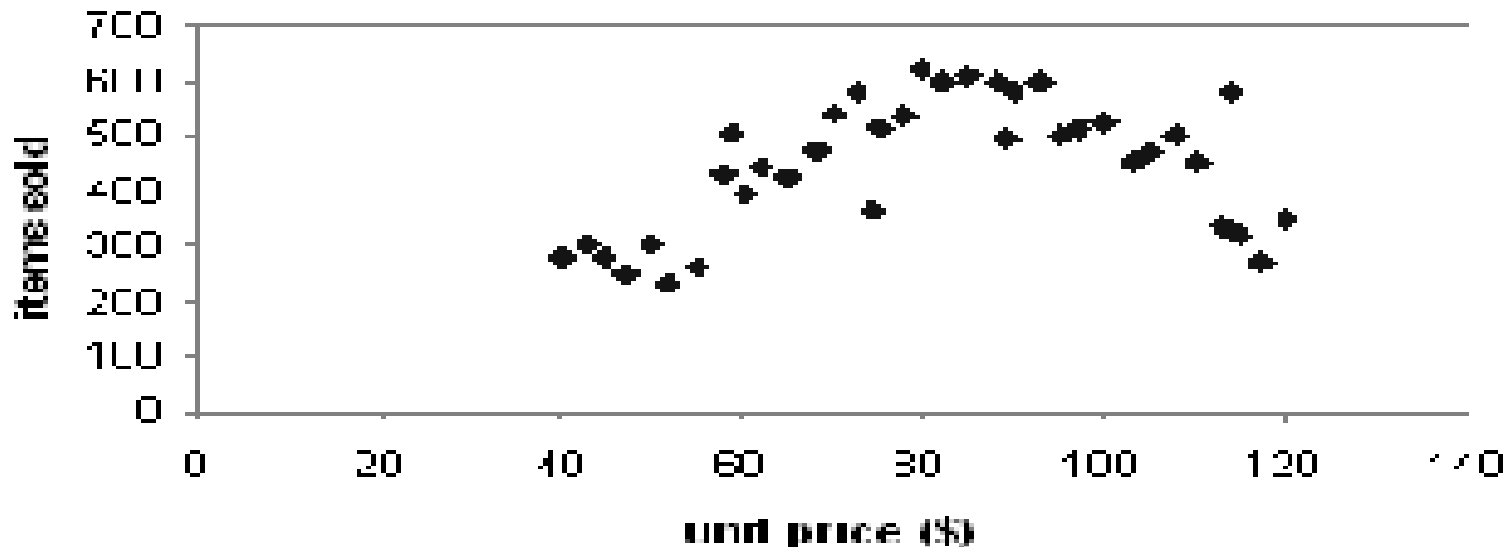
- There are **3 customers** waiting between **1 and 35 seconds**
- There are **5 customers** waiting between **1 and 40 seconds**
- There are **5 customers** waiting between **1 and 45 seconds**
- There are **5 customers** waiting between **1 and 50 seconds**
- There are **2 customers** waiting between **1 and 55 seconds**
- can conclude that the majority of customers wait between 35.1 and 50 seconds.

Customer Wait Time in Seconds (n=20)	
43.1	42.2
35.6	45.5
37.6	30.3
36.5	31.4
45.3	35.6
43.5	45.2
40.3	54.1
50.2	45.6
47.3	36.5
31.2	43.1



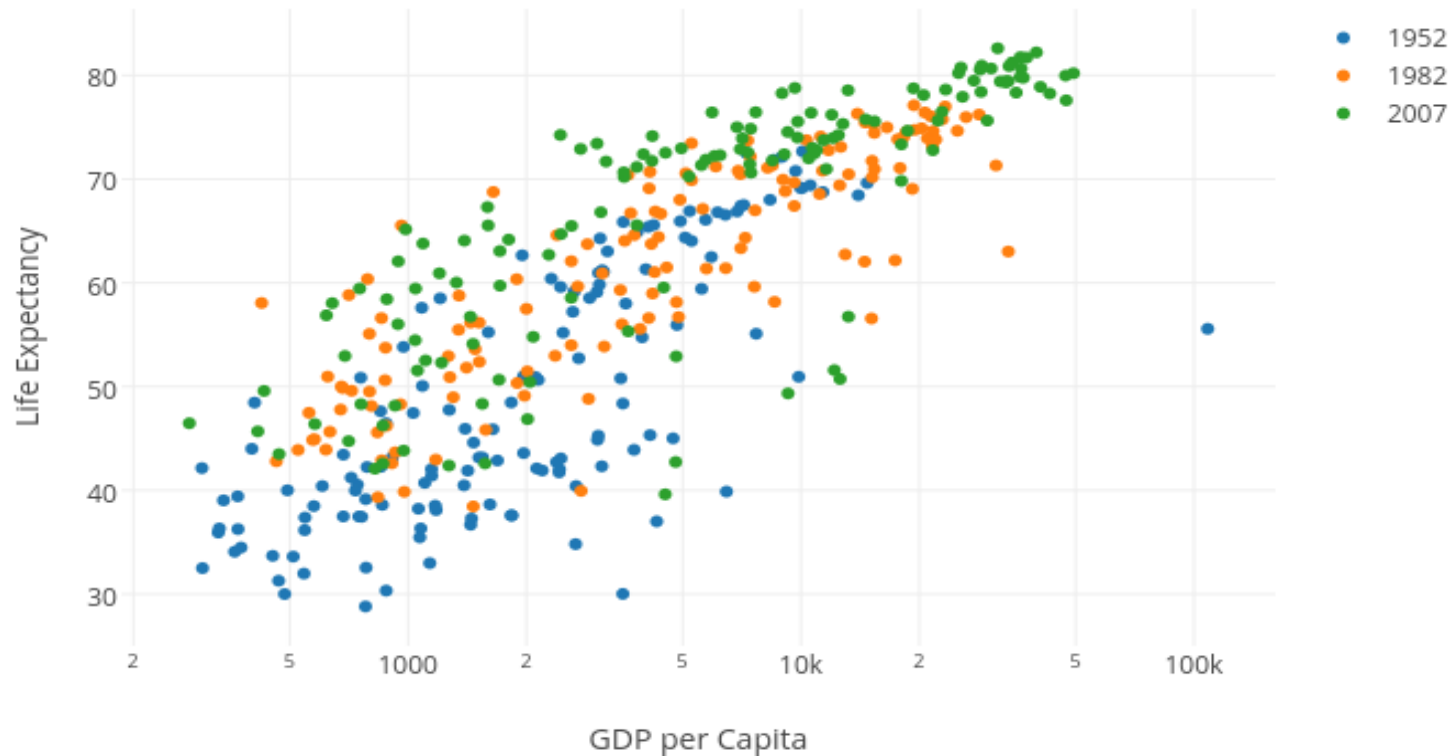
Scatter plot

- A scatter plot - effective graphical methods for determining if there appears to be a relationship, pattern, or trend between two numeric attributes.
- Provides a first look at bivariate data to see clusters of points, outliers, etc
- Each pair of values is treated as a pair of coordinates and plotted as points in the plane

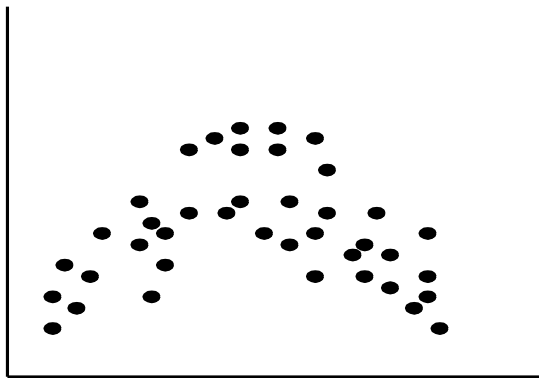
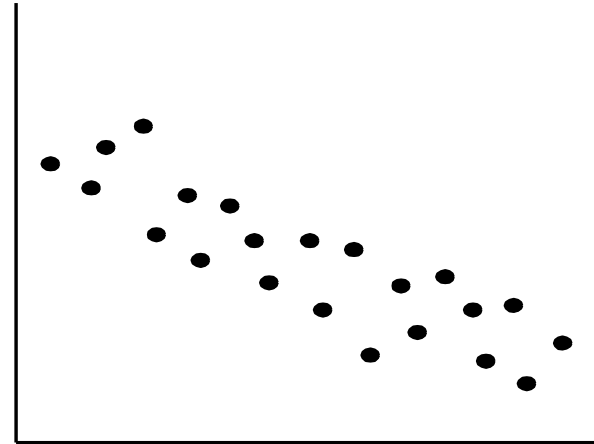
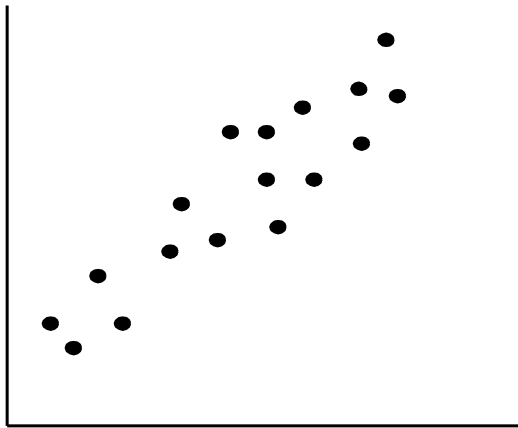


Scatter plot

- Provides a first look at bivariate data to see clusters of points, outliers, etc.
- Each pair of values is treated as a pair of coordinates and plotted as points in the plane

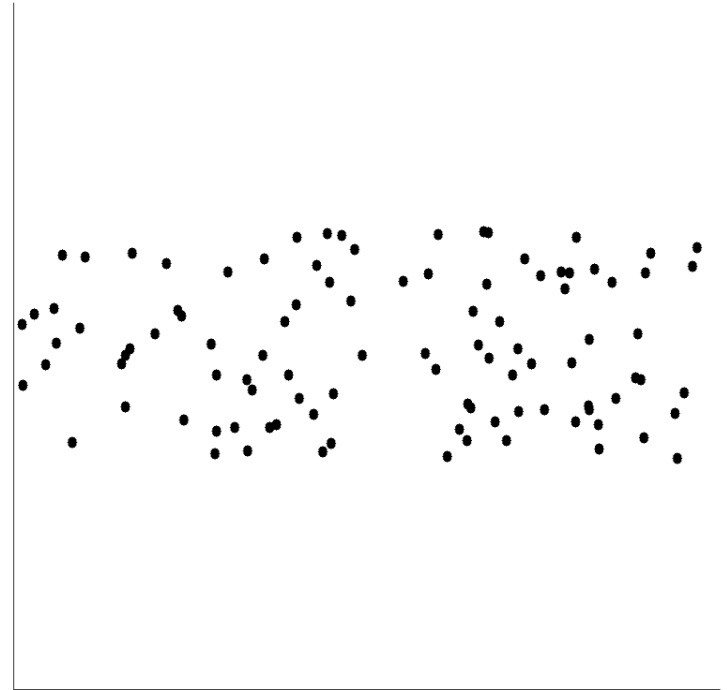
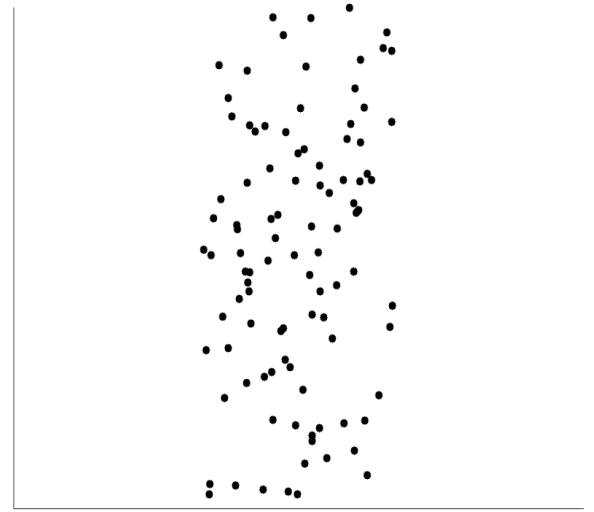
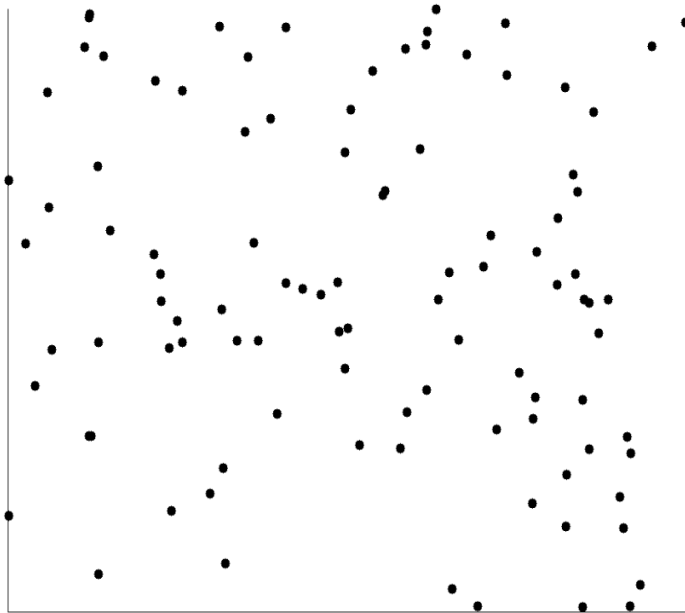


Positively and Negatively Correlated Data



- The left half fragment is positively correlated
- The right half is negative correlated

Uncorrelated Data



Rescaling Data

- Features are basically your column names and the respective data in that column will be of similar feature, this is in your everyday conventional datasets, most of them are usually in different quantitative measurements and in different magnitudes.
- Example the column with the name height will have data in cm (centimetre) and column with weight will have data in Kg(kilogram).
- Scaling data is the **process of increasing or decreasing the magnitude according** to a fixed ratio, in simpler words **you change the size but not the shape of the data** .

MinMax Scaler

- Scaling each feature to a given range
- This transformation does not change the distribution of the feature.
- Feature range parameter (default at (0,1)), Consider following example

Age = [44.9, 35.1, 28.2, 19.4, 28.9, 33.5, 22.0, 21.7, 30.9, 27.9]

Height = [70.4, 61.7, 75.3, 66.8, 66.9, 61.3, 61.7, 74.4, 76.5, 60.7]

$$X_{norm} = \frac{X - \min(X)}{\max(X) - \min(X)}$$

$$X_{norm} = \frac{X - \min(X)}{\max(X) - \min(X)} * (NewMax - NewMin) + NewMin$$

MinMax Scaler

Age = [44.9, 35.1, 28.2, 19.4, 28.9, 33.5, 22.0, 21.7, 30.9, 27.9]

Height = [70.4, 61.7, 75.3, 66.8, 66.9, 61.3, 61.7, 74.4, 76.5, 60.7]

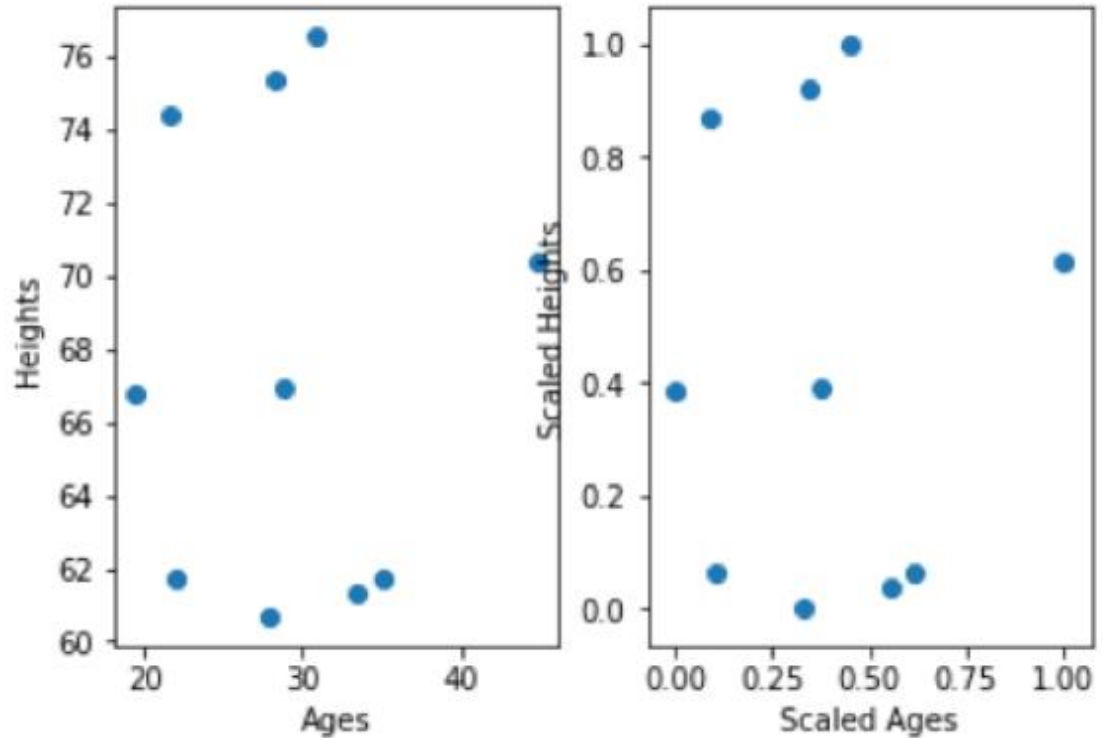
After re-scaling

Age = [1, 0.61568627, 0.34509804, 0, 0.37254902, 0.55294118,
0.10196078, 0.09019608, 0.45098039, 0.33333333]

Height = [0.61392405, 0.06329114, 0.92405063, 0.38607595,
0.39240506, 0.03797468, 0.06329114, 0.86708861, 1, 0]

MinMax Scaler

The same distribution of data, but rescaled in such a way that distances between points won't be biased by differences in scale



- Neural networks often require their inputs to be bounded between 0 and 1.
- In images, for example, where pixels can only take on a specific range of RGB values, data may have to be normalised.

MinMax Scaler

```
from sklearn.preprocessing import MinMaxScaler
import numpy as np
a=np.array([[70.4], [61.7], [75.3], [66.8], [66.9], [61.3],
scaler = MinMaxScaler(feature_range=(0, 1))
X_train_minmax = scaler.fit_transform(a)
X_train_minmax
```

```
array([[0.61392405],
       [0.06329114],
       [0.92405063],
       [0.38607595],
       [0.39240506],
       [0.03797468],
       [0.06329114],
       [0.86708861],
       [1.          ],
       [0.          ]])
```

STANDARDIZE Z-Score

- A Z-Score describes how far a raw score falls from the mean in the standard deviation units
 - Subtract the mean from the raw score
 - Divide by the standard deviation

$$z = +1.32$$

$$z = +0.01$$

$$z = -1.96$$

$$z = -1.43$$

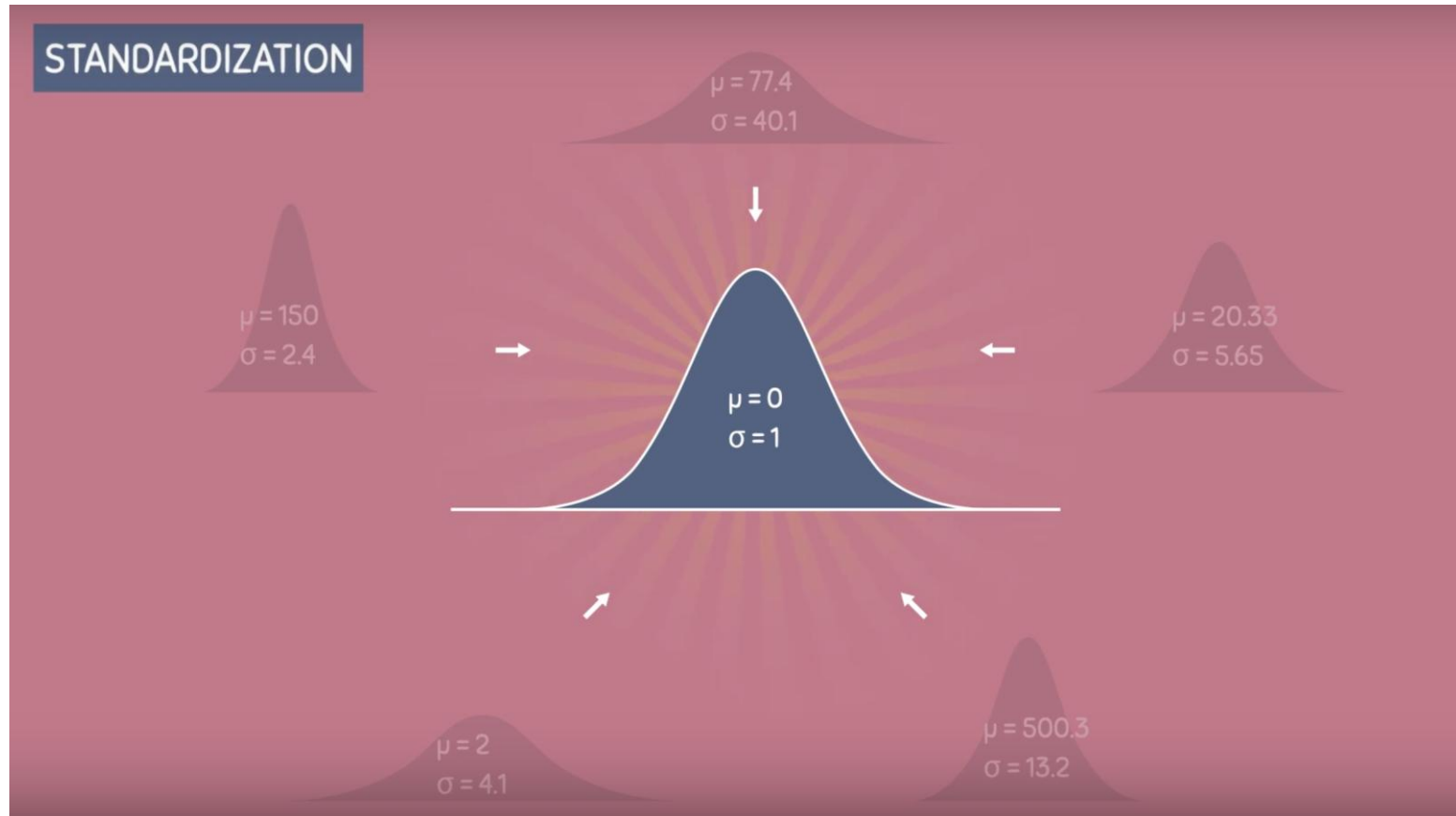
$$z = -2.22$$

$$z = +0.75$$

STANDARDIZE Z-Score

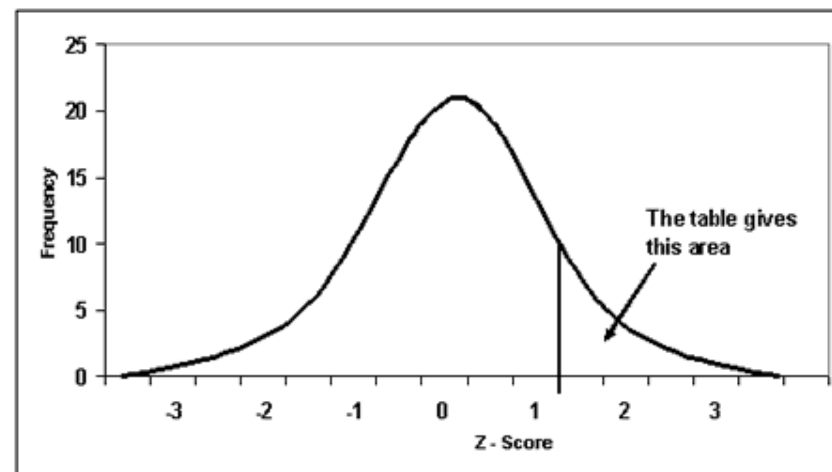
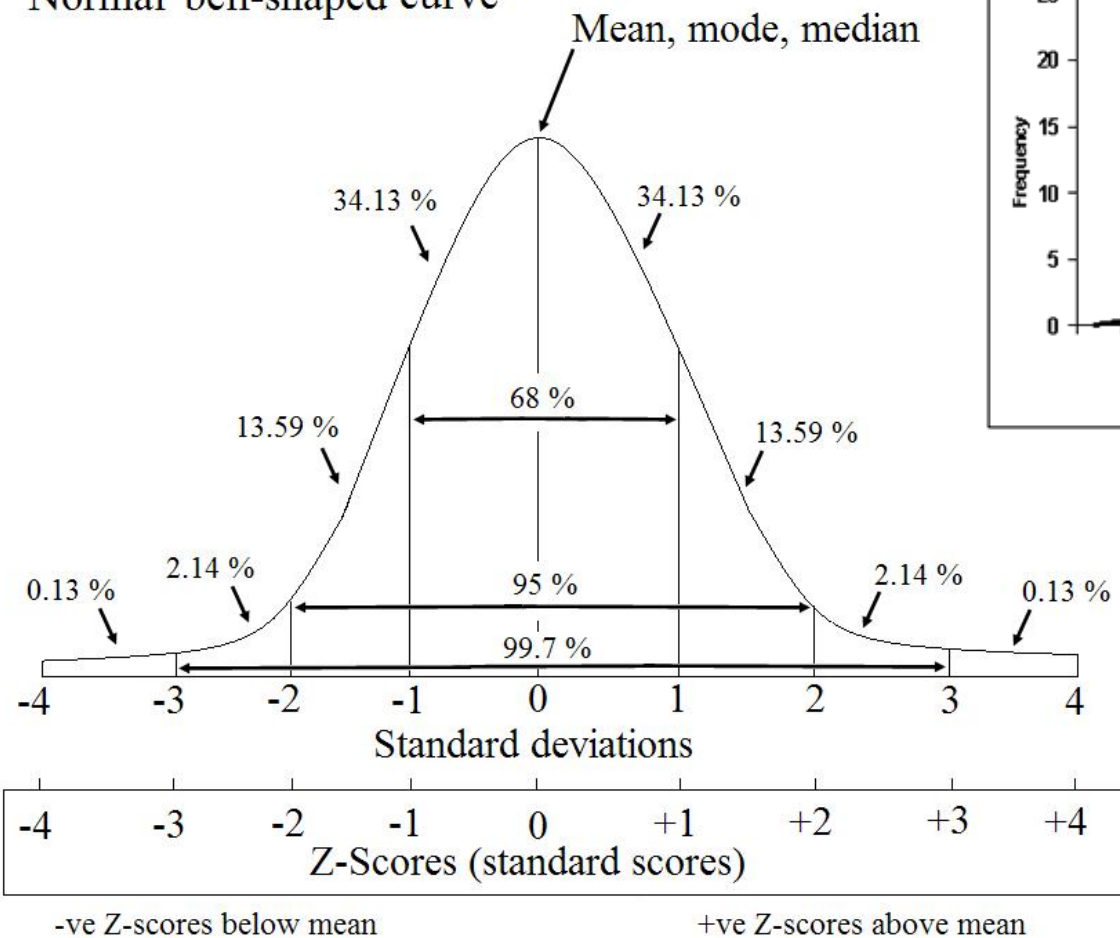
- Comparing scores from different distributions
- For computing relative frequency of scores from any distributions
- A standardised Z score always in the form of normal curve
- It has mean of 0 and standard deviation of 1
- Z score exceeding ± 1.96 are statically significant

STANDARDIZE Z-Score



STANDARDIZE Z-Score

Normal 'bell-shaped' curve



$$Z = \frac{\text{Score} - \text{Mean}}{\text{SD}}$$

Score

Mean

SD

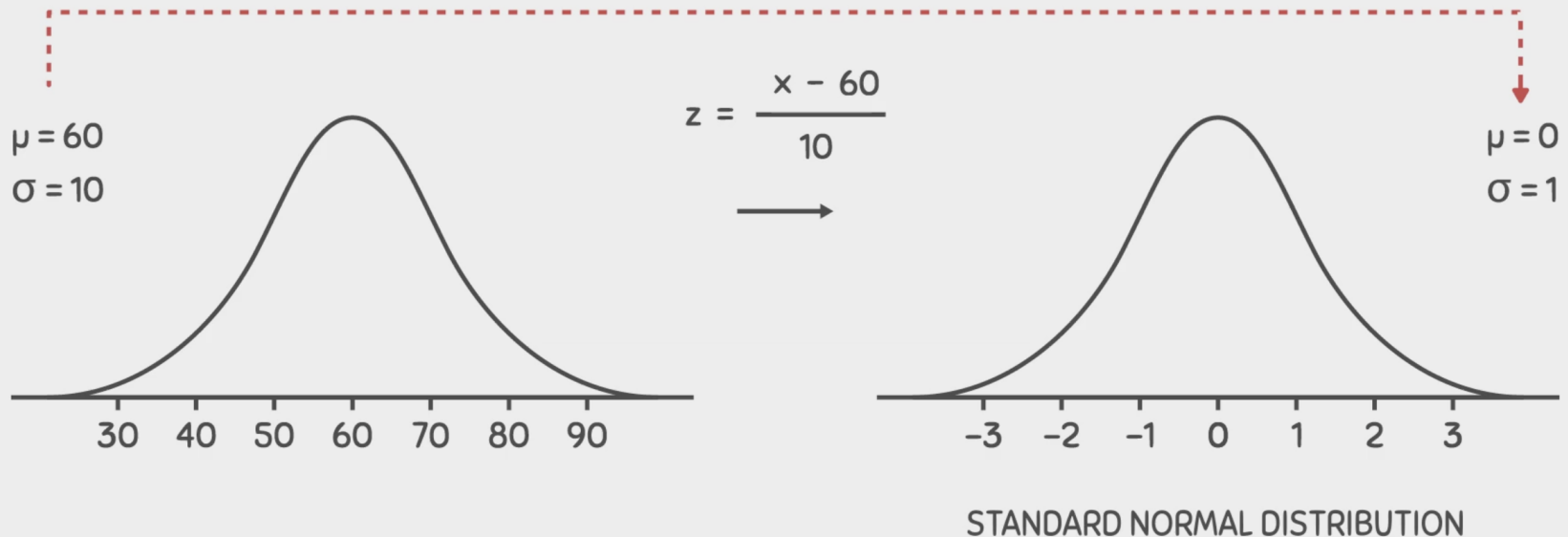
Z-Score - An example

- For instance, suppose you went to college in New York and your best friend went to college in Georgia.
- You might get a grade of 87 in a test with a mean of 77 and a standard deviation of 5, and the same day your friend might get a grade of 612 (mean 600, standard deviation 100).
- Although the two grades (87 and 612) can't be compared directly, the standardized values will allow you to immediately see who is doing better compared with the rest of the class.
- $(612 - 600) / 100 = 0.12$, so your friend's z-score or standardized grade is 0.12. $(87 - 77) / 5 = 2$, your standardized grade.
- With standardized data you have grounds to boast to your friend that you are doing much better than he or she is in the class

STANDARDIZE Z-Score

EXAMPLE

Suppose that we gathered data from last year's final chemistry exam and found that it followed a normal distribution with a mean of 60 and a standard deviation of 10.



Normalize or Standardize

- Normalization is good to use when you know that the distribution of your data does not follow a Gaussian distribution. This can be useful in algorithms that do not assume any distribution of the data like K-Nearest Neighbors and Neural Networks.
- Standardization, on the other hand, can be helpful in cases where the data follows a Gaussian distribution. However, this does not have to be necessarily true. Also, unlike normalization, standardization does not have a bounding range. So, even if you have outliers in your data, they will not be affected by standardization.

Binarize Data

- All values above the threshold are marked 1 and all equal to or below are marked as 0.
- This is called binarizing your data or threshold your data. It can be useful when you have probabilities that you want to make crisp values.
- The process of thresholding numerical features to get Boolean values
- It plays a key role in the discretization of continuous feature values.

	A	B	C	D
1	Country	Age	Salary	Purchased
2	France	44	72000	0
3	Spain	27	48000	1
4	Germany	30	54000	0
5	Spain	38	61000	0
6	Germany	40	1000	1
7	France	35	58000	1
8	Spain	78	52000	0
9	France	48	79000	1
10	Germany	50	83000	0
11	France	37	67000	1

Example :

A continuous data of pixels values of an 8-bit grayscale image have values ranging between 0 (black) and 255 (white) and one needs it to be black and white. So, using `Binarizer()` one can set a threshold converting pixel values from 0 – 127 to 0 and 128 – 255 as 1.

```
# For age, let threshold be 35
# For salary, let threshold be 61000
binarizer_1 = Binarizer(35)
binarizer_2 = Binarizer(61000)
```

One Hot Encoding

- To process categorical features, which can be in numerical or text form, they must be first transformed into a numerical representation.

Label Encoding

- The first column is the color column, which is all text
- Convert this kind of categorical text data into model-understandable numerical data, we use the Label Encoder
- After applying the **LabelEncoder**, → you'll see that the three colours in the first column have been replaced by the numbers 0, 1, and 2

	color	make	year
0	green	Chevrolet	2017
1	blue	BMW	2015
2	yellow	Lexus	2018

	color	make	year	color_encoded	make_encoded
0	green	Chevrolet	2017	1	1
1	blue	BMW	2015	0	0
2	yellow	Lexus	2018	2	2

The problem here is, since there are different numbers in the same column, the model will misunderstand the data to be in some kind of order, $0 < 1 < 2$. But this isn't the case at all. To overcome this problem, we use One Hot Encoder.

Labeled Encoding

A	0
B	1
C	2

Feature 1	Encoded Feature 1
A	0
A	0
A	0
A	0
B	1
B	1
B	1
C	2
C	2

One Hot Encoding

- It takes a column which has categorical data, which has been label encoded, and then splits the column into multiple columns
- For k distinct values, we can transform the feature into a k-dimensional vector with one value of 1 and 0 as the rest values.

	color	make	year	color_encoded	make_encoded	Color_0	Color_1	\
0	green	Chevrolet	2017	1	1	0.0	1.0	
1	blue	BMW	2015	0	0	1.0	0.0	
2	yellow	Lexus	2018	2	2	0.0	0.0	

	Color_2	Make0	Make1	Make2
0	0.0	0.0	1.0	0.0
1	0.0	1.0	0.0	0.0
2	1.0	0.0	0.0	1.0

One Hot Encoding

Label Binarizer

- The label binarizer class to perform one hot encoding in a single step

```
from sklearn.preprocessing import  
LabelBinarizer  
color_lb = LabelBinarizer()  
make_lb = LabelBinarizer()  
X =  
color_lb.fit_transform(df.color.val  
ues)  
Xm =  
make_lb.fit_transform(df.make.val  
ues)
```

To convert from the one-hot encoded vector back into the original text category, the label binarizer class provides the inverse transform function

```
print(X)  
green_ohe = X[[0]]  
color_lb.inverse_transform(g  
reen_ohe)
```



Any
Queries?